Exercise 1.

Show that axiom (M0) in the definition of a metric space is not needed, i.e. assuming that axioms (M1) to (M4) hold, prove that (M0) has to hold.

Solution. Putting $z = x$ into (M4) we obtain that $d(x,x) \leqslant d(x,y) + d(y,x)$, for all $x, y \in X$. Using (M2) and (M3) implies that $0 \leqslant 2d(x,y) \implies d(x,y) \geqslant 0$, for all $x, y \in X$. ∎

Exercise 2.

Prove the triangle inequality (M4) for Euclidean n-space:

$$d_2(x,z) \leqslant d_2(x,y) + d_2(y,z), \quad \forall x, y, z \in \mathbf{R}^n,$$

by first letting $r_i := x_i - y_i$ and $s_i := y_i - z_i$. So we need to prove

$$\Big(\sum_{i=1}^n (r_i + s_i)^2\Big)^{\frac{1}{2}} \leqslant \Big(\sum_{i=1}^n r_i^2\Big)^{\frac{1}{2}} + \Big(\sum_{i=1}^n s_i^2\Big)^{\frac{1}{2}}.$$

First show that this is equivalent to **Cauchy's Inequality**:

$$\Big(\sum_{i=1}^n r_i s_i\Big)^2 \leqslant \sum_{i=1}^n r_i^2 \sum_{i=1}^n s_i^2.$$

Now prove Cauchy's Inequality. *Hint: You may want to consider that, for all* $\lambda \in \mathbf{R}$, $\sum_{i=1}^n (r_i + \lambda s_i)^2 \geqslant 0$.

Solution. To obtain Cauchy's Inequality, square both sides of the equation to get

$$\sum_{i=1}^n (r_i + s_i)^2 = \sum_{i=1}^n r_i^2 + 2\sum_{i=1}^n r_i s_i + \sum_{i=1}^n s_i^2 \leqslant \sum_{i=1}^n r_i^2 + 2\Big(\sum_{i=1}^n r_i^2\Big)^{\frac{1}{2}}\Big(\sum_{i=1}^n s_i^2\Big)^{\frac{1}{2}} + \sum_{i=1}^n s_i^2,$$

which yields the result. To prove Cauchy's Inequality notice that, by the hint,

$$f(\lambda) := \lambda^2 \sum_{i=1}^n s_i^2 + 2\lambda \sum_{i=1}^n r_i s_i + \sum_{i=1}^n r_i^2 \geqslant 0.$$

Which implies that the quadratic equation $f(\lambda) = 0$ cannot have two distinct real roots. So the "discriminant" $b^2 - 4ac$ must be non-positive which gives the result.

For an alternative proof of Cauchy's Inequality, one can use the Cauchy–Schwarz inequality for the standard dot product on $\mathbf{R}^n$, see MTH107 Advanced Linear Algebra. ∎

Exercise 3.

Show that the following function defines a metric on $\mathbf{R}^n$:

$$d_\infty(x,y) := \max\{|x_1 - y_1|, \dots, |x_n - y_n|\}.$$

Solution. The axioms (M0) obviously holds. For (M1) and (M2), $d_\infty(x,y) = 0$ if and only if $|x_i - y_i| = 0$ for all $1 \leqslant i \leqslant n$, if and only if $x = y \in \mathbf{R}^n$. (M3) follows from $|x-y| = |y-x|$. For (M4), for each $1 \leqslant i \leqslant n$,

$$|x_i - z_i| \leqslant |x_i - y_i| + |y_i - z_i| \leqslant d_\infty(x,y) + d_\infty(y,z),$$

so $d_\infty(x,z) \leqslant d_\infty(x,y) + d_\infty(y,z)$. ∎

Exercise 4.

Determine whether or not the following functions d define metrics on the indicated sets:

a. $d(x,y) = |x^2 - y^2|$, for $x, y \in \mathbf{R}$;

b. $d(x,y) = |x^2 - y^2|$, for $x, y \in [0,\infty) \subseteq \mathbf{R}$;

c. $d(x,y) = (x_1 - y_1)^2 + (x_2 - y_2)^2$, for $x, y \in \mathbf{R}^2$.

Solution.

a. No, since $d(x,-x) = 0$ for $x \neq 0$ which goes against (M2).

b. Yes. For the triangle inequality $|x^2 - z^2| \leqslant |x^2 - y^2| + |y^2 - z^2|$ for all $x, y, z \geqslant 0$.

c. No. Notice that

$$4 = d((2,0),(0,0)) > d((2,0),(1,0)) + d((1,0),(0,0)) = 1 + 1 = 2;$$

so the triangle inequality fails. ∎

Exercise 5.

Show that the *British Railway*[a] metric on $\mathbf{R}^2$, given by

$$d(x,y) := \begin{cases} d_2(x,y), & \text{if } x \text{ and } y \text{ are linearly dependent;} \\ d_2(x,0) + d_2(0,y), & \text{otherwise,} \end{cases}$$

satisfies the axioms of a metric space, where d_2 is the Euclidean metric on $\mathbf{R}^2$.

[a] Also known as the French Railway metric.

Solution. The axioms (M0)–(M3) are easy to check. For (M4), first notice that $d_2(x,y) \leqslant d_2(x,0) + d_2(0,y)$, for all $x, y \in \mathbf{R}^2$. If x and z are linearly dependent then

$$d(x,z) = d_2(x,z) \leqslant d_2(x,y) + d_2(y,z) \leqslant d(x,y) + d(y,z).$$

If x and z are linearly independent, then either $\{x,y\}$ and $\{y,z\}$ are both linearly independent sets or y is dependent on either x or z. In the first case, (M4) is equivalent to $d_2(0,y) \geqslant 0$, which always holds, and in the second case (if we assume that x and y are dependent) then

$$d(x,z) = d_2(x,0) + d_2(0,z) \leqslant d_2(x,y) + d_2(y,0) + d_2(0,z) = d(x,y) + d(y,z).$$ ∎

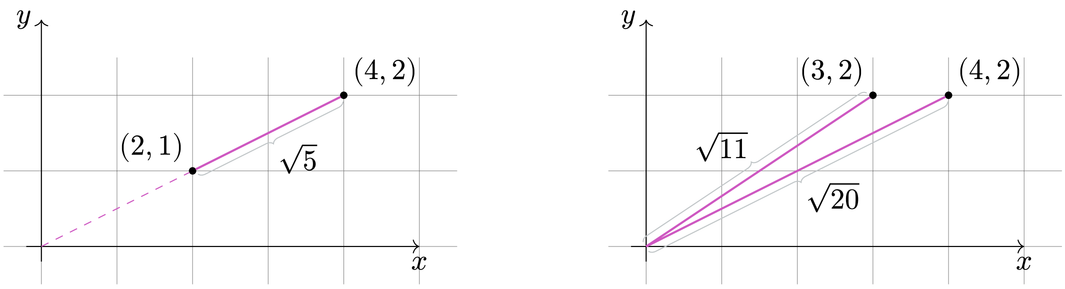


Figure 1: Shortest paths for the British railway distance between two points in the plane. We have $d((2,1),(4,2)) = \sqrt{5}$ while $d((3,1),(4,2)) = \sqrt{11} + \sqrt{20}$.

Exercise 6.

Given three points $x, y, z \in X$ in a metric space (X, d), show that

$$|d(x,z) - d(y,z)| \leqslant d(x,y).$$

Solution. It's equivalent to $-d(x,y) \leqslant d(x,z) - d(y,z) \leqslant d(x,y)$ and each inequality is equivalent to the triangle inequality. ∎

Exercise 7.

Show that the following functions between metric spaces are continuous.

a. The identity function: $\mathrm{Id}_X\colon (X,d) \to (X,d)$, where $\mathrm{Id}_X(x) := x$, for all $x \in X$.

b. Inclusion: $i\colon (W, d|_W) \to (X,d)$, where $W \subseteq X$ and $i(w) := w$, for all $w \in W$.

c. Constant function: $f\colon (X, d_X) \to (Y, d_Y)$, where $f(x) := y_0$, for all $x \in X$ and some fixed $y_0 \in Y$.

d. Show that every isometry is continuous.

Solution. a. Take $\delta = \varepsilon$.

b. Take $\delta = \varepsilon$.

c. Take any $\delta > 0$.

d. Since $d_Y(f(x), f(a)) = d_X(x,a)$ we can take $\delta = \varepsilon$. ∎

Exercise 8.

Suppose that $f\colon (X, d_X) \to (Y, d_Y)$ and $g\colon (Y, d_Y) \to (Z, d_Z)$ are continuous functions. Show that the composition

$$g \circ f\colon (X, d_X) \to (Z, d_Z),$$

where $(g \circ f)(x) := g(f(x))$, for all $x \in X$, is continuous.

Solution. Let $\varepsilon > 0$ and $a \in X$. We need to find $\delta > 0$ such that

$$d_X(x, a) < \delta \implies d_Z(g(f(x)), g(f(a))) < \varepsilon.$$

Since g is continuous at $f(a) \in Y$ there exists $\delta_g > 0$ such that

$$d_Y(y, f(a)) < \delta_g \implies d_Z(g(y), g(f(a))) < \varepsilon.$$

Also, f is continuous at $a \in X$ so there exists $\delta_f > 0$ such that

$$d_X(x, a) < \delta_f \implies d_Y(f(x), f(a)) < \delta_g.$$

We can then take $\delta = \delta_f$ to get what we need by setting $y = f(x)$. ■

Exercise 9.

Let (X, d_X) be a metric space and fix $x_0 \in X$. Show that the function

$$f\colon (X, d_X) \implies (\mathbf{R}, d_{\mathbf{R}}),$$

given by $f(x) := d_X(x, x_0)$, for all $x \in X$, is continuous (where $d_{\mathbf{R}}$ is the usual metric on $\mathbf{R}$).

Solution. Let $\varepsilon > 0$ and $a \in X$. Then we need to find $\delta > 0$ such that $d_X(x, a) < \delta \implies |f(x) - f(a)| < \varepsilon$. This is equivalent to

$$d_X(a, x_0) - \varepsilon < d_X(x, x_0) < d_X(a, x_0) + \varepsilon.$$

Since for any $x \in X$, $d_X(x, x_0) \leqslant d_X(x, a) + d_X(a, x_0)$ and $d_X(a, x_0) \leqslant d_X(a, x) + d_X(x, x_0)$ we can just take $\delta = \varepsilon$. ■

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

a. Compute $d_{\sup}(x^2, x)$ in $\mathscr{C}[0,1]$ and $d_1(x^2, x)$ in $\mathcal{L}_1[0,1]$.

b. Assume that $1 < a < b$ in $\mathbf{R}$. Compute $d_{\sup}(x^3, 2-x^2)$ in $\mathscr{C}[a,b]$ and $d_1(x^3, 2-x^2)$ in $\mathcal{L}_1[a,b]$.

Solution. a. On $[0,1]$ we have $|x^2-x| = x-x^2$. Moreover, $\frac{\mathrm{d}(x-x^2)}{\mathrm{d}x} = 1-2x$ is 0 if and only if $x = \frac{1}{2}$. So $x = \frac{1}{2}$ is a local extremum. Moreover $\frac{\mathrm{d}^2(x-x^2)}{\mathrm{d}x^2} = -2 < 0$ when $x = \frac{1}{2}$, so $x = \frac{1}{2}$ is a local maximum. We conclude $d_{\sup}(x^2, x) = \sup_{x\in[0,1]}(x-x^2) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ in $\mathscr{C}[0,1]$.

For d_1 a direct computation gives us $d_1(x^2, x) = \int_0^1 (x-x^2)\,\mathrm{d}x = [\frac{x^2}{2} - \frac{x^3}{3}]_0^1 = \frac{1}{2} - \frac{1}{3} - 0 = \frac{1}{6}$ in $\mathcal{L}_1[0,1]$.

b. Since $1 < a < b$, we have $x^3 + x^2 > 2$ on $[a,b]$ and it attains its absolute maximum at $x = b$. So $d_{\sup}(x^3, 2-x^2) = b^3 + b^2 - 2$ in $\mathscr{C}[a,b]$. In $\mathcal{L}_1[a,b]$

$$d_1(x^3, 2-x^2) = \int_a^b |x^3 + x^2 - 2|\,\mathrm{d}x = \left[\frac{x^4}{4} + \frac{x^3}{3-2x}\right]_a^b,$$

which reduces to $\frac{b^4-a^4}{4} + \frac{b^3-a^3}{3} - 2(b-a)$. ■

Exercise 2.

Let $X := \{0,1\}^\infty$ be the set of all infinite **binary sequences** $x = x_0x_1x_2\ldots$, where $x_n \in \{0,1\}$, for $n \geqslant 0$; so $x = 100110111\ldots$ is a typical element. Show that the following are metrics on X.

a.
$$d_{\min}(x,y) := \begin{cases} 0, & \text{if } x = y; \\ \frac{1}{2^n}, & \text{if } n = \min\{m \mid x_m \neq y_m\}. \end{cases}$$

b.
$$d^*(x,y) := \sum_{j=0}^{\infty} \frac{|x_j - y_j|}{2^j}.$$

Find two elements $x, y \in X$ such that $d_{\min}(x,y) \neq d^*(x,y)$.

Solution. a. (M1)–(M3) are satisfied automatically. For (M4), let $d_{\min}(x,y) = \frac{1}{2^p}$ and $d_{\min}(y,z) = \frac{1}{2^q}$. So $d_{\min}(x,z) = \frac{1}{2^r}$, where $r = \min\{p,q\}$ if $p \neq q$ and $r > \min\{p,q\}$ if $p = q$. In both cases $\frac{1}{2^r} \leqslant \frac{1}{2^p} + \frac{1}{2^q}$, as required. If $x = y$ or $y = z$, the proof is trivial.

b. Notice that the right hand side is bounded above by 2 and therefore converges. The axioms (M1)–(M4) then follow by elementary properties of convergent series.

Taking $x = 0100\ldots$ and $y = 100\ldots$ gives $d_{\text{min}}(x, y) = 1$ and $d^*(x, y) = \frac{3}{2}$. ■

Exercise 3.

Explain why neither of the following functions define a metric on the set of continuous functions $[0, 1] \to \mathbf{R}$:

a. $d(f, g) = |f(\frac{1}{2}) - g(\frac{1}{2})|$;

b. $d(f, g) = \sup_{x\in[0,1]}|f'(x) - g'(x)|$, where $f'(x) := \frac{\mathrm{d}f(x)}{\mathrm{d}x}$.

Solution. a. Let $f(x) = x$ and $g(x) = 1 - x$. Then $d(f, g) = 0$ so axiom (M2) fails.

b. If f or g fails to be differentiable for some $x \in [0, 1]$, then $d(f, g)$ is not defined! But even so, the examples $f(x) = 1$ and $g(x) = 0$ show that $d(f, g) = 0$ so (M2) fails again. ■

Exercise 4.

Let $\mathcal{X}$ be the space of bounded sequences of real numbers. Show that the map $\|\cdot\|_\infty : \mathcal{X} \to \mathbf{R}$ defined by $\|a\|_\infty := \sup_{n\in\mathbf{N}}|a_n|$ is a norm on $\mathcal{X}$.

Solution. Before verifying the norm axioms, we need to check that the map $\|\cdot\|_\infty$ is well-defined. This is the case as a bounded implies that the supremum exists (and belongs to $\mathbf{R}$).

"(N0)" It directly follows from the definition that $\|a\|_\infty \geqslant 0$ for all a.

"(N1) and (N2)" $\|a\|_\infty = 0$ if and only if $|a_n| = 0$ for all n, if and only if a is the zero sequence.

"(N3)" $\|\lambda a\|_\infty = \sup_{n\in\mathbf{N}}|\lambda a_n| = \sup_{n\in\mathbf{N}}|\lambda||a_n| = |\lambda|\sup_{n\in\mathbf{N}}|a_n| = |\lambda|\|\lambda a\|_\infty$.

"(N4)" Let a and b be two sequences of real numbers. For every n we have $|a_n + b_n| \leqslant |a_n| + |b_n|$. This implies that for bounded sequences $\|a + b\|_\infty = \sup_{n\in\mathbf{N}}|a_n + b_n| \leqslant \sup_{n\in\mathbf{N}}|a_n| + \sup_{n\in\mathbf{N}}|b_n| = \|a\|_\infty + \|b\|_\infty$. ■

Exercise 5.

Let $\mathcal{Y}$ be the space of absolutely convergent sequences of real numbers. Show that the map $\|\cdot\|_1 : \mathcal{Y} \to \mathbf{R}$ defined by $\|a\|_1 := \sum_{n\in\mathbf{N}}|a_n|$ is a norm on $\mathcal{Y}$. (You may need to revise your properties of absolutely convergent series.)

Solution. By definition, $\mathcal{Y}$ is the space of sequences a such that $\|a\|_1$ is well-defined.

"(N0)–(N2)" These are straightforward verifications, similar to Question 4

Before going further, let us recall the Rearrangement Theorem: The terms of an absolutely convergent series can be rearranged in any order, and the series will still converge to the same sum.

"(N3)" $\|\lambda a\|_1 = \sum_{n\in\mathbf{N}} |\lambda a_n| = \sum_{n\in\mathbf{N}} |\lambda||a_n| = |\lambda| \sum_{n\in\mathbf{N}} |a_n| = |\lambda|\|\lambda a\|_1$.

"(N4)" Once again, for two sequences a and b of real numbers we have $|a_n + b_n| \leqslant |a_n| + |b_n|$ for every n. This implies that for absolutely convergent sequences $\|a+b\|_1 = \sum_{n\in\mathbf{N}} |a_n + b_n| \leqslant \sum_{n\in\mathbf{N}} (|a_n| + |b_n|) = \sum_{n\in\mathbf{N}} |a_n| + \sum_{n\in\mathbf{N}} |b_n| = \|a\|_\infty + \|b\|_\infty$. ∎

Exercise 6.

Show that for any norm $\|\cdot\|$ we have $\|x \pm y\| \geqslant |\|x\| - \|y\||$.

Solution. This is equivalent to

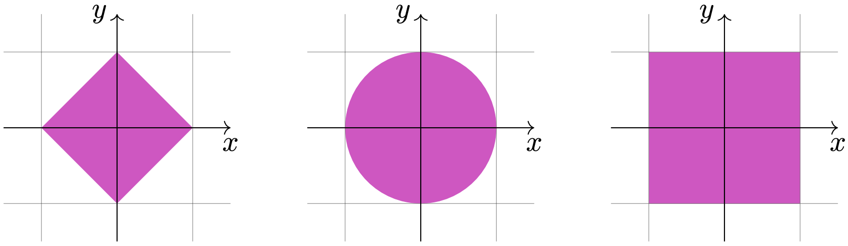
$$-\|x \pm y\| \leqslant \|x\| - \|y\| \leqslant \|x \pm y\|.$$

The right inequality is equivalent to $\|x\| - \|y\| \leqslant \|x \pm y\|$. But this follows from the triangle inequality applied to $x = (x \pm y) \mp y$ and the fact that $\|-y\| = \|y\|$. The proof of the left inequality is similar. ∎

Exercise 7.

Sketch open balls of radius 1 around the origin in $\mathbf{R}^2$ with the different metrics d_1, d_2 and d_∞.

Solution.



∎

For d_1, the ball consist of all points (x, y) such that $|x + y| < 1$. The equation $|x + y| = 1$ gives us 4 straight segments (depending on the positivity of x and y) between the points $(0, 1)$, $(1, 0)$, $(0, -1)$ and $(-1, 0)$. The open ball is the interior of the square delimited by these segments.

For d_2, the equation $1 = d((x, y), 0) = \sqrt{x^2 + y^2}$ defines a circle of radius 1.

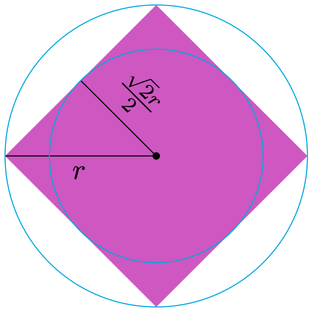
For d_∞, the equation $1 = d((x, y), 0) = \max(x, y)$ defines a square with corners $(\pm 1, \pm 1)$.

Exercise 8.

Let $B_r(x)$ and $B'_r(x)$ denote open balls of radius r in $\mathbf{R}^2$ with respect to the Euclidean metric and the taxicab metric respectively. Find real numbers s and t such that

$$B_1(x) \subseteq B'_s(x) \subseteq B_{\sqrt{2}}(x) \subseteq B'_t(x) \subseteq B_2(x).$$

Solution. Each open ball $B'_r(x)$ has bounding line of greatest Euclidean distance r from x and least distance $r\frac{\sqrt{2}}{2} = \frac{r}{\sqrt{2}}$.



Therefore,

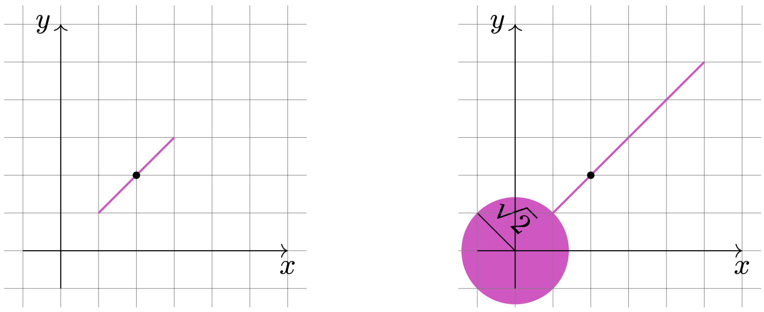
$$1 \leqslant \frac{s}{\sqrt{2}} < s \leqslant \sqrt{2} \leqslant \frac{t}{\sqrt{2}} < t \leqslant 2.$$

So the only possible solutions are $s = \sqrt{2}$ and $t = 2$. ■

Exercise 9.

Sketch open balls of radius $\sqrt{2}$ and $3\sqrt{2}$ around the point $(2, 2)$ in $\mathbf{R}^2$ with respect to the British Railway metric (see Tutorial 1, Question 5).

Solution.



■

If the radius r of the ball is smaller than the Euclidean distance from the center c to 0, then the balls consists of points p that are linearly dependent with c and of distance at most r to c. This is simply a line segment (on the line between 0 and c) of length $2r$ centred at c.

If $r \geqslant d_2(c, 0)$, then to the above line segment we can add $B_{r-d_2(c,0)}(0; d_2)$. Indeed, once we have reach 0 we can go back in any direction, but not further away than $r - d_2(c, 0)$.

Exercise 10.

Prove that if $x, y \in (X, d)$ are points in a metric space with $d(x, y) = 2r > 0$, then $B_r(x)$ and $B_r(y)$ are disjoint. Is it possible that in a metric space X containing more than one point that the only open sets are $\varnothing$ and X?

Solution. Since $d(x, y) > 0$ we know that $x \neq y$. Suppose that $z \in B_r(x) \cap B_r(y)$. Then $d(x, y) \leqslant d(x, z) + d(z, y) < 2r$, which contradicts $d(x, y) = 2r$.

It is not possible since if $x, y \in X$ with $x \neq y$, then $d(x, y) = 2r > 0$, for some $r \in \mathbf{R}$. Then we know that $B_r(x) \cap B_r(y) = \varnothing$ and both of these open balls are non-empty and open in X. ∎

To go further

The same idea shows that if X is a metric space containing at least n points, then it has at least n open sets different from $\varnothing$ and X.

***Exercise 11.**

Let X be the set of all closed intervals $[a, b]$ of $\mathbf{R}$. Show that

$$d([a, b], [c, d]) := \max\{|c - a|, |d - b|\}$$

is a metric on X.

Now consider the metric space (H, d_∞), where

$$H := \{(x_1, x_2) \in \mathbf{R}^2 \mid x_1 \leqslant x_2\},$$

and show that it is isometric to (X, d).

Solution. The axioms (M1) and (M2) are satisfied because $|c - a| = |d - b| = 0$ if and only if $c = a$ and $d = b$. Axiom (M3) is satisfied because $|x - y| = |y - x|$, for all $x, y \in \mathbf{R}$. For (M4), consider intervals $I = [a, b]$, $J = [c, d]$ and $K = [e, f]$ in X. Then

$$d(I, J) = \max\{|c - a|, |d - b|\} \leqslant \max\{|c - e| + |e - a|, |d - f| + |f - b|\},$$

since $|x - z| \leqslant |x - y| + |y - z|$, for all $x, y, z \in \mathbf{R}$. Also, $|c - e| + |e - a|$ and $|d - f| + |f - b|$ are both less than or equal to $\max\{|e - a|, |f - b|\} + \max\{|e - c|, |f - d|\}$ so

$$d(I, J) \leqslant \max\{|e - a|, |f - b|\} + \max\{|e - c|, |f - d|\} = d(I, K) + d(K, J).$$

We define a function $f \colon X \to H$ by $f([a, b]) = (a, b)$. It is easy to see that f is a bijection. Also, since

$$d_\infty((x_1, x_2), (y_1, y_2)) = \max\{|y_1 - x_1|, |y_2 - x_2|\},$$

f gives an isometry between (X, d) and (H, d_∞). ∎

Exercise 1.

Given any two subsets $U, V \subseteq (X, d)$ of a metric space show that

a. $U \subseteq V \implies U^\circ \subseteq V^\circ$;

b. $(U^\circ)^\circ = U^\circ$;

c. U° is open in X;

d. U° is the largest subset of U that is open in X.

Solution. a. If $u \in U^\circ$, then $\exists \varepsilon > 0$ such that $B_\varepsilon(u) \subseteq U \subseteq V$; so $u \in V^\circ$;

b. By definition $(U^\circ)^\circ \subseteq U^\circ$. On the other hand, if $u \in U^\circ$, then $\exists \varepsilon > 0$ such that $B_\varepsilon(u) \subseteq U$, so $B_\varepsilon(u)^\circ \subseteq U^\circ$ by part a, but $B_\varepsilon(u)$ is open by Lemma 1.5.12 and so $B_\varepsilon(u) = B_\varepsilon(u)^\circ$. Therefore, $u \in (U^\circ)^\circ$, so $U^\circ \subseteq (U^\circ)^\circ$.

c. This follows immediately from part b.

d. Suppose that $U^\circ \subseteq W \subseteq U$, where W is open in X. Then $W = W^\circ \subseteq U^\circ$ by part a. So $W \subseteq U^\circ$, and $U^\circ = W$. ■

Exercise 2.

Show that an open subset U of a metric space is a union of open balls.

Solution. Since U is open, for every $u \in U$, $\exists \varepsilon_u > 0$ such that $B_{\varepsilon_u}(u) \subseteq U$. Therefore, $\bigcup_{u \in U} B_{\varepsilon_u}(u) \subseteq U$. And since $u \in B_{\varepsilon_u}(u)$ we also have $U \subseteq \bigcup_{u \in U} B_{\varepsilon_u}(u)$. ■

Exercise 3.

Consider the set of binary sequences $\{0,1\}^\infty$ with the metric d^* from Tutorial 2, Question 2. What is the maximum possible value of $d^*(x, y)$, and for which pairs of sequences is this attained?
Describe the elements of the open ball $B_{3/4}(0)$ in $\{0,1\}^\infty$, where 0 denotes the zero sequence $00\dots$.

Solution. The maximum possible value of $d^*(x, y)$ is $\sum_{j=0}^\infty \frac{1}{2^j} = 2$. It is attained by any pair of sequences which differ in every element; for example $d^*(0, 11\dots) = 2$.

By definition, $B_{\frac{3}{4}}(0)$ contains all x such that $\sum_{j=0}^\infty \frac{1}{2^j} < \frac{3}{4}$. Thus $x_0 = 0$, otherwise $d^*(x, 0) \geqslant 1$. If $x_0 = 0$ and $x_1 = 1$, then $d^*(x, 0) = \frac{1}{2} + \sum_{j=2}^\infty \frac{1}{2^j}$, so it is necessary that $x_2 = 0$ and $x_3 x_4 \dots \neq 11\dots$. On the other hand, if $x_0 = x_1 = 0$, then $d^*(x, 0) \leqslant \frac{1}{2}$.

So $x \in B_{\frac{3}{4}}(0)$ if and only if $x = 00z$ for any sequence z, or $x = 010w$ for any sequence $w \neq 11\dots$. The additional elements in $\overline{B}_{\frac{3}{4}}(0)$ are $01011\dots$ and $01100\dots$. ■

Exercise 4.

On $\mathbf{R}^n$ check that

$$d_\infty(x,y) \leqslant d_2(x,y) \leqslant d_1(x,y) \leqslant n d_\infty(x,y).$$

where d_1 is the taxicab metric, d_2 is the Euclidean metric, and d_∞ is the max metric.

Solution. The first and second inequalities follow from the definitions; you may find them easier to deduce by first considering the following observation:

$$(d_\infty(x,y))^2 \leqslant (d_2(x,y))^2 \leqslant (d_1(x,y))^2.$$

The final inequality follows directly from the definitions. ∎

Exercise 5.

Let (X, d_X) and (Y, d_Y) be two metric spaces and consider one of the metrics d_a, d_b, d_c on $X \times Y$.

a. Prove that the projection maps:

$$p_1 : X \times Y \to X \quad \text{and} \quad p_2 : X \times Y \to Y$$

given by $p_1(x,y) = x$ and $p_2(x,y) = y$ are continuous.

b. For some fixed $x_0 \in X$ and $y_0 \in Y$ prove that

$$i_{y_0} : X \to X \times Y \quad \text{and} \quad i_{x_0} : Y \to X \times Y$$

given by $i_{y_0}(x) = (x, y_0)$ and $i_{x_0}(y) = (x_0, y)$ are continuous.

Solution. Since d_a, d_b and d_c have the same open sets, continuity for d_a is equivalent to continuity for d_b, and also to continuity for d_c.

a. We will consider the metric d_a on $X \times Y$ and prove that p_1 is continuous. Let $(x,y) \in X \times Y$ and $\varepsilon > 0$. Then

$$d_X(p_1(x,y), p_1(x',y')) = d_X(x,x') \leqslant d_a((x,y),(x',y')) = d_X(x,x') + d_Y(y,y').$$

So we find that p_1 is continuous at $(x,y) \in X \times Y$ by taking $\delta = \varepsilon$. The proof for p_2 is symmetric.

b. We will prove that i_{y_0} is continuous taking the metric d_a again on $X \times Y$. Let $x \in X$ and $\varepsilon > 0$. Then

$$d_a(i_{y_0}(x), i_{y_0}(x')) = d_a((x,y_0),(x',y_0)) = d_X(x,x') + d_Y(y_0,y_0) = d_X(x,x').$$

So by taking $\delta = \varepsilon$ again we see that i_{y_0} is continuous at $x \in X$. The proof for p_2 is symmetric. ∎

Exercise 6.

Given a metric space (X, d), consider $(X \times X, d_a)$.

a. Prove that the diagonal map
$$\Delta \colon X \to X \times X$$
given by $\Delta(x) = (x, x)$ is continuous.

b. Show that the metric $d \colon (X \times X, d_a) \to (\mathbf{R}, d_{\mathbf{R}})$ is a continuous function.

Solution. a. Let $x \in X$ and $\varepsilon > 0$. Then
$$d_a(\Delta(x), \Delta(y)) = d_a((x, x), (y, y)) = d(x, y) + d(x, y) = 2d(x, y).$$
So we can take $\delta = \frac{1}{2}\varepsilon$ in the ε-δ definition of continuity.

b. Let $(x, y) \in X \times X$ and $\varepsilon > 0$. Then we need to find $\delta > 0$ such that
$$d_a((x, y), (x', y')) < \delta \implies |d(x, y) - d(x', y')| < \varepsilon.$$
Suppose for an instant that $d(x, y) \geqslant d(x', y')$. Then $d(x, y) - d(x', y') \geqslant 0$ and thus we have
$$\begin{aligned} |d(x, y) - d(x', y')| &= d(x, y) - d(x', y') \\ &\leqslant (d(x, x') + d(x', y') + d(y', y)) - d(x', y') \\ &= d(x, x') + d(y, y'). \end{aligned}$$
If $d(x, y) \leqslant d(x', y')$, then
$$\begin{aligned} |d(x, y) - d(x', y')| &= d(x', y') - d(x, y) \\ &\leqslant (d(x', y') + d(y', y) + d(y, x)) - d(x, y) \\ &= d(x, x') + d(y, y'). \end{aligned}$$
Since $d_a((x, y), (x', y')) = d(x, x') + d(y, y')$ we can take $\delta = \varepsilon$. ■

Exercise 7.

Suppose (X, d) is a metric space. Define $d'(x, y) = \min\{1, d(x, y)\}$. Prove that d' is a metric on X and then show that d and d' are topologically equivalent. Give an example to show that they may not be Lipschitz equivalent.

Solution. We have $d'(x, y) \leqslant d(x, y)$ and $0 < d'(x, y) \leqslant 1$, for all $x \neq y \in X$. So for the triangle inequality notice that
$$\begin{aligned} d'(x, z) = \min\{1, d(x, z)\} &\leqslant \min\{1, d(x, y) + d(y, z)\} \\ &\leqslant \min\{1, d(x, y)\} + \min\{1, d(y, z)\} = d'(x, y) + d'(y, z). \end{aligned}$$

The other axioms of a metric are straightforward to check.

In terms of open balls, notice that

$$B_r(x;d') = \begin{cases} B_r(x;d), & \text{if } r \leqslant 1; \\ X, & \text{if } r > 1, \end{cases}$$

so $B_r(x;d) \subseteq B_r(x;d')$, for all $x \in X$ and $r > 0$. Therefore, it automatically follows that if $U \subseteq X$ is d'-open, then it is d-open. Suppose now that $U \subseteq X$ is d-open and let $u \in U$. Then $\exists \varepsilon > 0$ such that $B_\varepsilon(u;d) \subseteq U$. Let $\varepsilon' := \min\{1, \varepsilon\} \leqslant 1$. Then $B_{\varepsilon'}(u;d') \subseteq B_\varepsilon(u;d) \subseteq U$ so U is d'-open.

Obviously, for any metric space (X,d), we find that (X,d') is bounded. But an unbounded metric space cannot be Lipschitz equivalent to a bounded one by Lemma 1.10.5. For a concrete counterexample, take $X = \mathbf{R}$ and $d = d_{\mathbf{R}}$ its standard metric. If d and d' were Lipschitz equivalent, then there would exist $h, k \in (0, \infty)$ such that

$$hd'(x,y) \leqslant d(x,y) = |x-y| \leqslant kd'(x,y) \leqslant k, \quad \forall x, y \in \mathbf{R}.$$

Taking x and y to be further than distance k away from each other, for example, $x = 0, y = k+1$, we find a contradiction. ■

Exercise 8.

Prove that isometries and Lipschitz equivalences between metric spaces are continuous.

Solution. Isometries are continuous by taking $\delta = \varepsilon$ in the definition of continuity. A Lipschitz equivalence $f\colon (X, d_X) \to (Y, d_Y)$ (using the same notation as in its definition) is continuous at $x \in X$ by taking $\delta = h\varepsilon$ since:

$$d_Y(f(x), f(y)) \leqslant \frac{1}{h} d_X(x,y) < \frac{\delta}{h} = \varepsilon. \qquad ■$$

Exercise 9.

Check that being an isometry defines an equivalence relation on all metric spaces. Do the same for Lipschitz equivalences and homeomorphisms.

Solution. "$X \sim X$" The identity map $\mathrm{Id}\colon X \to X$ is bijective and an isometry as well as being its own inverse. It is thus a Lipschitz equivalence and a continuous map.

"$X \sim Y \Rightarrow Y \sim X$" If $f\colon X \to Y$ is an isometry, respectively a homeomorphism, then $f^{-1}\colon Y \to X$ is automatically an isometry, respectively a homeomorphism. If f is a Lipschitz equivalence, with parameters h and k, then

$$hd_Y(f(x), f(y)) \leqslant d_X(x,y) \leqslant kd_Y(f(x), f(y)).$$

By letting $w = f(x)$ and $z = f(y)$ we have

$$hd_Y(w,z) \leqslant d_X(f^{-1}(w), f^{-1}(y)) \leqslant kd_Y(w,z),$$

which is equivalent to

$$\frac{1}{k} d_X(f^{-1}(w), f^{-1}(y)) \leqslant d_Y(w, z) \leqslant \frac{1}{h} d_Y(w, z).$$

So f^{-1} is also a Lipschitz equivalence (with parameters $\frac{1}{k}$ and $\frac{1}{h}$).

"$X \sim Y, Y \sim Z \Rightarrow X \sim Z$" This follows directly from the composition of continuous functions. ∎

Exercise 10.

In the proof that two Lipschitz equivalent metrics d and d' on X are topologically equivalent (Proposition 1.8.6) prove the statements regarding open balls.
That is, show that if f is a (h, k)-Lipschitz equivalence, then $f(B_{h\varepsilon}(x)) \subseteq B_\varepsilon(f(x))$ and $f^{-1}(B_{\frac{\varepsilon}{k}}(y)) \subseteq B_\varepsilon(f^{-1}(y))$.

Solution. Let $y \in B_{h\varepsilon}(x; d)$. Then $d(x, y) < h\varepsilon$, which implies that $d'(f(x), f(y)) \leqslant \frac{1}{h} d(x, y) < \varepsilon$. This proves that $f(y) \in B_\varepsilon(f(x))$ for any $y \in B_{h\varepsilon}(x)$.

The formula for f^{-1} follows from the above formula and the fact that f^{-1} is a $(\frac{1}{k}, \frac{1}{h})$-Lipschitz equivalence. ∎

Exercise 1.

Prove that the **open annulus** $\{x \in \mathbf{R}^2 \mid 4 < x_1^2 + x_2^2 < 9\}$ and the **punctured plane** $\mathbf{R}^2 \setminus \{0\}$ are homeomorphic subspaces of $\mathbf{R}^2$.

Solution. The annulus $A := \{x \in \mathbf{R}^2 \,|\, 4 < x_1^2 + x_2^2 < 9\}$ lies between the circles of radius 2 and 3, so every $a \in A$ may be written as $(r\cos\theta, r\sin\theta)$ for some $r \in (2,3) \subseteq \mathbf{R}$. By Example 1.8.5, there exists a homoeomorphism $t\colon (2,3) \to (0,+\infty)$ (for example, one can take $t(x) = \tan(\frac{\pi}{2} \cdot x)$). Define

$$h\colon A \to \mathbf{R}^2 \setminus \{0\} \quad \text{by} \quad h(a) = (t(r)\cos\theta, t(r)\sin\theta).$$

For a given θ, both $r\cos\theta \mapsto t(r)\cos\theta$ and $r\sin\theta \mapsto t(r)\sin\theta$ are continuous function from $(2,3)$ to $\mathbf{R}$. It follows that h itself is continuous. Also, h has a continuous inverse $h^{-1}(s\cos\theta, s\sin\theta) = (t^{-1}(s)\cos\theta, t^{-1}(s)\sin\theta)$. For our concrete example, one has $t^{-1}(s) = \frac{2}{\pi}\tan^{-1} s$. ■

Exercise 2.

Let $\ell^1 = (\mathcal{Y}, d_1)$ be the sequence space of absolutely convergent sequences (see Definition 1.4.3) with the d_1 metric.
Find a distance preserving injection of the space $(\{0,1\}^\infty, d^*)$ (see Tutorial 3, Question 4) into ℓ^1.
Find an example of a point $a \in \mathcal{Y}$ such that $a \in \overline{B}_{\frac{1}{10}}(0)$ but $a \notin B_{\frac{1}{10}}(0)$.

Solution. Consider the function $i\colon \{0,1\}^\infty \to \mathcal{Y}$ defined by

$$i(x_0 x_1 \dots x_k \dots) = (x_0, \frac{x_1}{2}, \dots, \frac{x_k}{2^k}, \dots);$$

the right hand side is well defined, because

$$\sum_{k\in\mathbf{N}} \left|\frac{x_k}{2^k}\right| \leqslant \sum_{k\in\mathbf{N}} \frac{1}{2^k} = 2,$$

as required. Obviously i is injective; it also preserves distance, because

$$d^*(x,y) = \sum_{k\in\mathbf{N}} \frac{|x_k - y_k|}{2^k} = \sum_{k\in\mathbf{N}} \left|\frac{x_k}{2^k} - \frac{y_k}{2^k}\right| = d_1(i(x), i(y)).$$

A point $a \in \mathcal{Y}$ satisfies the condition if and only if $\sum_{k=0}^{\infty} |a_k| = \frac{1}{10}$. Any example, such as

$$\left(\frac{1}{10}, 0, 0, \dots\right) \quad \text{or} \quad \left(\frac{1}{20}, \frac{1}{40}, \dots, \frac{1}{10\cdot 2^{j+1}}, \dots\right)$$

will do. ■

Exercise 3.

Given any two subsets $U, V \subseteq X$ prove the following:

a. $U \subseteq V$ implies $\overline{U} \subseteq \overline{V}$;

b. $\overline{\overline{V}} = \overline{V}$;

c. $\overline{V}$ is closed in X;

d. $\overline{V}$ is the smallest set containing V that is closed in X.

Solution. a. For every $\varepsilon > 0$, if $B_\varepsilon(x) \cap U$ is non-empty, so is $B_\varepsilon(x) \cap V$.

b. By definition, $\overline{V} \subseteq \overline{\overline{V}}$. Conversely, if $x \in \overline{\overline{V}}$, then $B_{\frac{\varepsilon}{2}}(x) \cap \overline{V}$ is non-empty for every $\varepsilon > 0$. So there exists w for which $d(x, w) < \frac{\varepsilon}{2}$ and $w \in \overline{V}$, which implies that $B_{\frac{\varepsilon}{2}}(w) \cap V$ is non-empty; therefore $B_\varepsilon(x) \cap V$ is also non-empty, by the triangle inequality. Hence $x \in \overline{V}$ and $\overline{\overline{V}} \subseteq \overline{V}$ as required.

c. This follows immediately from b.

d. Suppose that $V \subseteq W$, where W is closed. Then $\overline{V} \subseteq \overline{W} = W$ by a. So $\overline{V} \subseteq W$, showing the minimality of $\overline{V}$. ■

Exercise 4.

Determine whether the following subsets are open, closed, or neither in $\mathbf{R}$ (with the standard metric):

a. $[0, 1)$;

b. $\mathbf{Z}$;

c. $\{x \in \mathbf{R} \mid \sin x > 0\}$;

d. $\bigcup_{n=2}^{\infty} [\frac{1}{n}, 1)$.

Solution. a. $[0, 1)$ is not open since 0 does not lie in the interior. It is not closed because 1 is a closure point that does not belong to the set.

b. $\mathbf{Z}$ is closed in $\mathbf{R}$ because its complement is open as a union of open intervals.

c. $\{x \in \mathbf{R} \mid \sin x > 0\} = \bigcup_{n \in \mathbf{Z}} (2n\pi, (2n + 1)\pi)$ is a union of open intervals in $\mathbf{R}$, and is therefore open.

d. $\bigcup_{n=2}^{\infty} [\frac{1}{n}, 1) = (0, 1)$ is open in $\mathbf{R}$. ■

Exercise 5.

For a subset $W \subseteq (X, d)$ of a metric space prove that

a. $W^\circ = X \setminus (\overline{X \setminus W})$;

b. $\overline{W} = X \setminus (X \setminus W)^\circ$.

Solution. a. $w \in W^\circ \iff \exists \varepsilon > 0 : B_\varepsilon(w) \subseteq W \iff \exists \varepsilon > 0 : B_\varepsilon(w) \cap (X \setminus W) = \varnothing \iff w \notin \overline{X \setminus W}$.

b. $x \in \overline{W} \iff \forall \varepsilon > 0 : B_\varepsilon(x) \cap W \neq \varnothing \iff \forall \varepsilon > 0 : B_\varepsilon(x) \nsubseteq X \setminus W \iff x \notin (X \setminus W)^\circ$. ■

Exercise 6.

In a metric space (X, d) prove that every open ball is contained in a closed ball and that every closed ball is contained in an open ball. Is it always true that $\overline{B_r(x)}$ is $\overline{B}_r(x)$? Give a proof or a counterexample.

Solution. For any $x \in X$ and $r > 0$,

$$B_r(x) \subseteq \overline{B}_r(x) \subseteq B_{2r}(x).$$

It is not true that the closure of $B_r(x)$ is always $\overline{B}_r(x)$. In a discrete metric space (X, d) we the ball $B_1(x) = \{x\}$ is both open and close, so $\overline{B_1(x)} = \{x\}$ as well. But $\overline{B}_1(x) = X$, which is not the same as $\{x\}$ if X contains more than one point. ■

Exercise 7.

Consider a subset $W \subseteq (X, d)$ of a metric space. Recall that a point $x \in X$ is an accumulation point of W if, for every $\varepsilon > 0$, $B_\varepsilon(x) \cap W \setminus \{x\} \neq \varnothing$. Let $\mathrm{Acc}(W)$ denote the set of all accumulation points of W. Prove that $\overline{W} = W \cup \mathrm{Acc}(W)$.
Recall that $w \in W$ is an isolated point of W if there exists $\varepsilon > 0$ such that $B_\varepsilon(w) \cap W = \{w\}$. Let $\mathrm{Iso}(W)$ denote the set of all isolated points of W. Prove that $\mathrm{Iso}(W) = W \setminus \mathrm{Acc}(W)$.

Solution. Let $x \in \overline{W}$. Then if $x \notin W$ we have have that $B_\varepsilon(x) \cap W \setminus \{x\} \neq \varnothing$, for every $\varepsilon > 0$. So $x \in \mathrm{Acc}(W)$. It is easy to see that if $x \in \mathrm{Acc}(W)$, then $x \in \overline{W}$.

Let $w \in \mathrm{Iso}(W)$. Then $w \in W$ by definition and $B_\varepsilon(w) \cap W = \{w\}$, for some $\varepsilon > 0$, which implies that $B_\varepsilon(w) \cap W \setminus \{w\} = \varnothing$ so $w \notin \mathrm{Acc}(W)$. Suppose that $w \in W \setminus \mathrm{Acc}(W)$. Then there is an $\varepsilon > 0$ such that $B_\varepsilon(w) \cap W \setminus \{w\} = \varnothing$ which implies that $B_\varepsilon(w) \cap W = \{w\}$ so $w \in \mathrm{Iso}(W)$. ■

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Prove that if $Y \subseteq Z$, where Z is a bounded subset of (X, d), then Y is bounded and $\operatorname{diam} Y \leqslant \operatorname{diam} Z$.

Solution. Since Z is bounded we know there exists $x \in X$ and $r > 0$ such that $Z \subseteq \overline{B}_r(x)$ and since $Y \subseteq Z$ we have that Y is bounded. Also, for each $y_1, y_2 \in Y \subseteq Z$, $d(y_1, y_2) \leqslant \operatorname{diam} Z$ so $\operatorname{diam} Z$ is an upper bound for $\{d(y_1, y_2) \,|\, y_1, y_2 \in Y\}$ which implies that $\operatorname{diam} Y := \sup\{d(y_1, y_2) \,|\, y_1, y_2 \in Y\} \leqslant \operatorname{diam} Z$. ■

Exercise 2.

Prove that if Y, Z are bounded subsets of (X, d) with $Y \cap Z \neq \varnothing$, then

$$\operatorname{diam} Y \cup Z \leqslant \operatorname{diam} Y + \operatorname{diam} Z.$$

Solution. If $u_1, u_2 \in Y \cup Z$ both belong to Y, then

$$d(u_1, u_2) \leqslant \operatorname{diam} Y \leqslant \operatorname{diam} Y + \operatorname{diam} Z.$$

If u_1, u_2 both belong to Z, a interchanging the roles of Y and Z gives the desired result. If $u_1 \in Y$ and $u_2 \in Z$ (or the the other way around) then take some $\alpha \in Y \cap Z$ and

$$d(u_1, u_2) \leqslant d(u_1, \alpha) + d(\alpha, u_2) \leqslant \operatorname{diam} Y + \operatorname{diam} Z.$$

Therefore, $d(u_1, u_2) \leqslant \operatorname{diam} Y + \operatorname{diam} Z$, for all $u_1, u_2 \in Y \cup Z$, which gives the result. ■

Exercise 3.

For a subset $W \subseteq (X, d)$ of a metric space, prove that $\partial W = \overline{W} \cap \overline{(X \setminus W)}$.

Solution. $x \in \partial W \iff x \in \overline{W} \setminus W^\circ \iff (x \in \overline{W} \,\&\, \forall \varepsilon > 0 : B_\varepsilon(x) \cap (X \setminus W) \neq \varnothing) \iff (x \in \overline{W} \,\&\, x \in \overline{(X \setminus W)})$. ■

Exercise 4.

Determine ∂S, S°, and $\overline{S}$ for each of the subspaces $S := [0, 1) \cup (2, 3]$, $S := \mathbf{Z}$, and $S := \mathbf{R} \setminus \mathbf{Q}$ of the Euclidean line.

Solution.

$$\partial[0,1) \cup (2,3] = \{0,1,2,3\}, \quad ([0,1) \cup (2,3])^\circ = (0,1) \cup (2,3),$$
$$\overline{[0,1) \cup (2,3]} = [0,1] \cup [2,3].$$

$$\partial \mathbf{Z} = \mathbf{Z}, \quad (\mathbf{Z})^\circ = \varnothing, \quad \overline{\mathbf{Z}} = \mathbf{Z}.$$
$$\partial(\mathbf{R} \setminus \mathbf{Q}) = \mathbf{R}, \quad (\mathbf{R} \setminus \mathbf{Q})^\circ = \varnothing, \quad \overline{(\mathbf{R} \setminus \mathbf{Q})} = \mathbf{R}.$$ ■

Exercise 5.

Is $\mathbf{Q}^n$ dense in $\mathbf{R}^n$? Let $\mathbf{Z}[\frac{1}{2}] := \{\frac{m}{2^n} \mid m, n \in \mathbf{Z}\}$. Is $\mathbf{Z}[\frac{1}{2}]$ dense in $\mathbf{R}$? Is $\mathbf{Z}[\frac{1}{2}]^n$ dense in $\mathbf{R}^n$.

Solution. To prove that $\mathbf{Q}^n$ is dense in $\mathbf{R}^n$ we must prove that $\overline{\mathbf{Q}^n} = \mathbf{R}^n$. It is sufficient to show that every $(x_1, \dots, x_n) \in \mathbf{R}^n$ is the limit of a sequence in $\mathbf{Q}^n$. But we know that $\mathbf{Q}$ is dense in $\mathbf{R}$, so there exists a sequence $(q_{i,k})_{k\geqslant 1}$ converging to x_i for every $1 \leqslant i \leqslant n$. Then $\left(q_k = (q_{1,k}, \dots, q_{n,k})\right)_{k\geqslant}$ converges to x, because

$$d(x, q_k) = \left(\sum_{i=1}^{n} (x_i - q_{i,k})^2\right)^{\frac{1}{2}} \to 0, \quad \text{as } k \to \infty.$$

Given any $z \in \mathbf{R}$, let z_n denote the truncation of its binary expansion after n digits, for every $n \geqslant 1$. So $z_n \in \mathbf{Z}[\frac{1}{2}]$ by construction, and $z_n \to z$ as $n \to \infty$. So $\mathbf{Z}[\frac{1}{2}]$ is dense in $\mathbf{R}$. Using a similar argument as for $\mathbf{Q}^n$ shows that $\mathbf{Z}[\frac{1}{2}]^n$ is dense in $\mathbf{R}^n$. ∎

Exercise 6.

Show that a subset $U \subseteq X$ of a metric space is open if and only if for every sequence (x_n) of X that converges to a point in U, there exists $K \in \mathbf{N}$ such that $x_k \in U$, for all $k \geqslant K$.

Solution. "$\Rightarrow$" Suppose that U is open in X and that (x_n) is a sequence in X that converges to $u \in U$. Since U is open, there exists $\varepsilon > 0$ such that $B_\varepsilon(u) \subseteq U$, and for this $\varepsilon > 0$, $\exists K \in \mathbf{N}$ such that $k \geqslant K \implies d(x_k, u) < \varepsilon$, i.e. $x_k \in B_\varepsilon(u) \subseteq U$, for all $k \geqslant K$.

"$\Leftarrow$" Let $u \in U$ and suppose $u \notin U^\circ$, that is, for every $\varepsilon > 0$, $B_\varepsilon(u) \cap (X \setminus U) \neq \varnothing$. Choose $x_i \in B_{\frac{1}{i}}(u) \cap (X \setminus U)$. Then (x_n) is a sequence in X such that $\lim_n x_n = u$ but $x_k \notin U$, for all $k \in \mathbf{N}$. ∎

Exercise 7.

Let $\mathscr{P}[0,1] \subseteq \mathscr{C}[0,1]$ be the subspace consisting of all *polynomials functions*. By constructing the sequence

$$f_n(x) := 1 + \frac{x}{2} + \frac{x^2}{4} + \dots + \frac{x^n}{2^n} \in \mathscr{P}[0,1],$$

show that $\mathscr{P}[0,1]$ is not closed in $\mathscr{C}[0,1]$.

Solution. Let $f(x) = (1 - \frac{x}{2})^{-1}$ in $\mathscr{C}[0,1]$; then the Binomial Theorem implies that $f(x) - f_n(x) = \sum_{k\geqslant n+1} \frac{x^k}{2^k} \leqslant \frac{1}{2^n}$ for $x \in [0,1]$. So

$$d_{\sup}(f_n, f) = \sup_{x\in[0,1]} |f_n(x) - f(x)| = \frac{1}{2^n} \to 0, \quad \text{as } n \to \infty,$$

and $f_n \to f$ in $\mathscr{C}[0,1]$. But f is not a polynomial, so $\mathscr{P}[0,1]$ is not closed. ∎

Exercise 8.

Given the sequence space $(\{0,1\}^\infty, d^*)$ (see Tutorial 3 Question 4), consider $f_1 \colon \{0,1\}^\infty \to \{0,1\}^\infty$ defined by inserting 1 at the beginning of every sequence. Prove that f_1 is continuous and decreases the distance in the sense that $d^*(f_1(x), f_1(y)) \leqslant d^*(x,y)$.

Solution. By definition $f_1(x) = 1x_0x_1 \dots$ and $f_1(y) = 1y_0y_1 \dots$. So

$$d^*(f_1(x), f_1(y)) = \sum_{j=1}^{\infty} \frac{|x_{j-1} - y_{j-1}|}{2^j} = \frac{1}{2} \sum_{j-1=0}^{\infty} \frac{x_{j-1} - y_{j-1}}{2^{j-1}} = \frac{1}{2} d^*(x,y).$$

In particular, $d^*(f_1(x), f_1(y)) \leqslant d^*(x,y)$, with equality only if $x = y$.

It also follows from $d^*(f_1(x), f_1(y)) = \frac{1}{2} d^*(x,y)$that f_1 is continuous by choosing $\delta \leqslant 2\varepsilon$ in the ε-δ definition of continuity. ■

Exercise 9.

Show that if $A \subseteq \mathbf{R}^n$ is a bounded subset, then $A \subseteq [a,b]^n$ for some $a, b \in \mathbf{R}$.

Solution. Since A is bounded, it is contained in a closed ball $\overline{B}_r(x)$ for some point $x = (x_1, \dots, x_n) \in \mathbf{R}^n$ and some radius $r \in \mathbf{R}$. Let $a := \min\{x_i - r \mid i \in \{1, \dots, n\}\}$ and let $b := \max\{x_i + r \mid i \in \{1, \dots, n\}\}$. Then one easily verify that the ball $\overline{B}_r(x)$ is contained in $[a,b]^n$ and hence $A \subseteq \overline{B}_r(x) \subseteq [a,b]^n$ as desired. ■

Exercise 10.

Give examples of:

a. a nested sequence of non-empty closed sets $V_n \subseteq (0,1)$ with $\operatorname{diam} V_n \to 0$, as $n \to \infty$, but $\bigcap_{n \in \mathbf{N}} V_n = \varnothing$;

b. a nested sequence of non-empty sets U_n in $\mathbf{R}$ with $\operatorname{diam} U_n \to 0$, as $n \to \infty$, but $\bigcap_{n \in \mathbf{N}} U_n = \varnothing$;

c. a nested sequence of non-empty closed sets $W_n \subseteq \mathbf{R}$ with $\bigcap_{n \in \mathbf{N}} W_n = \varnothing$.

Solution. a. $(V_n) = \left((0, \frac{1}{n}]\right)_{n \geqslant 2}$;

b. $(U_n) = \left((0, \frac{1}{n})\right)_{n \geqslant 2}$;

c. $(W_n) = \left([n, +\infty)\right)_{n \geqslant 1}$. ■

***Exercise 11.**

Suppose that X is a compact metric space and $V_1 \supseteq V_2 \supseteq \cdots$ is a nested sequence of closed subsets of X. Prove that

$$\operatorname{diam}\left(\bigcap_{n=1}^{\infty} V_n\right) = \inf\{\operatorname{diam} V_n \mid n \in \mathbf{N}\}.$$

Solution. If, say, V_{n_0} is empty, then $\bigcap_{n=1}^{\infty} V_n = \varnothing$ and its diameter is 0 by definition. Likewise in this case $\operatorname{diam} V_{n_0} = 0$ so $\inf\{\operatorname{diam} V_n \mid n \in \mathbf{N}\} = 0$ also. Suppose now that all the V_n are non-empty. (We already know that their intersection is then non-empty.) Now $\bigcap_{n=1}^{\infty} V_n \subseteq V_m$ for any $m \in \mathbf{N}$ so $\operatorname{diam}\left(\bigcap_{n=1}^{\infty} V_n\right) \leqslant \operatorname{diam} V_m$. Hence $\operatorname{diam}\left(\bigcap_{n=1}^{\infty} V_n\right) \leqslant \inf\{\operatorname{diam} V_m \mid m \in \mathbf{N}\} = m_0$ say.

Conversely m_0 is a lower bound for the diameters of the V_n so for any $\varepsilon > 0$ and any $n \in \mathbf{N}$ we know that $\operatorname{diam} V_n > m_0 - \varepsilon$. Hence there exist points $x_n, y_n \in V_n$ such that $d(x_m, x_n) > m_0 - \varepsilon$. Since X is sequentially compact (x_n) has a subsequence (x_{n_r}) converging to a point $x \in X$, and then (y_{n_r}) has a subsequence $(y_{n_{r_s}})$ converging to a point $y \in X$. Since $(x_{n_{r_s}})$ is a subsequence of (x_{n_r}) it too converges to x. Also, by continuity of the metric, $d(x_{n_{r_s}}, y_{n_{r_s}}) \to d(x, y)$ as $s \to \infty$. Hence $d(x, y) \geqslant m_0 - \varepsilon$. Also, $x, y \in V_n$ for each $n \in \mathbf{N}$ since V_n is closed in X. Since this is true for all $n \in \mathbf{N}$ we have $x, y \in \bigcap_{n=1}^{\infty} V_n$. Hence $\operatorname{diam} \bigcap_{n=1}^{\infty} V_n \geqslant m_0 - \varepsilon$. But this is true for any $\varepsilon > 0$, so $\operatorname{diam} \bigcap_{n=1}^{\infty} V_n \geqslant m_0$. The above taken together proves the result. ■

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Which of the following subsets of $\mathbf{R}^2$ are connected? Which are path-connected?

a. $B_1((1,0)) \cup B_1((-1,0))$;

b. $\overline{B}_1((1,0)) \cup \overline{B_1((-1,0))}$;

c. $\overline{B}_1((1,0)) \cup B_1((-1,0))$;

d. The "rational comb" $C := \{(q,y) \in \mathbf{R}^2 \mid q \in \mathbf{Q}, y \in [0,1]\} \cup (\mathbf{R} \times \{1\})$;

e. The set of all points in $\mathbf{R}^2$ with at least one coordinate in $\mathbf{Q}$.

Solution. a. This is a disjoint union of two open sets. It is thus not connected and therefore not path-connected.

b. This is a union of two path-connected sets (the two balls) with non-empty intersection (it contains $(0,0)$). It is therefore path-connected, and hence connected.

c. The two balls are path-connected. For x in $B_1((-1,0))$ the line segment between 0 and x stays inside $\overline{B}_1((1,0)) \cup B_1((-1,0))$ and is thus a path from 0 to x. We conclude that $\overline{B}_1((1,0)) \cup B_1((-1,0))$ is path-connected, and hence connected.

d. A point of the form (q,y) $(q \in \mathbf{Q}, y \in [0,1])$ is connected by a (vertical) path to $(q,1)$, which is connected by a (horizontal path) to $(0,1)$. All points of the form $(r,1)$ are connected by a path to $(0,1)$. We conclude that C is path-connected, and hence connected.

e. Any point (p,r) with $p \in \mathbf{Q}$ is connected by a vertical path to $(p,0)$, which is connected to $(0,0)$ by a horizontal path. Similarly, any point (r,q) with $q \in \mathbf{Q}$ is path-connected. We conclude that our set is path-connected, and hence connected. ■

Exercise 2.

Suppose that $A \mid B$ is a partition of a metric space X and that $f\colon X \to Y$ is a function to another metric space Y. Prove that if the restrictions $f|_A$ and $f|_B$ are both continuous, then f is continuous.

Solution. Let U be an open subset of Y. Set $U_A := f^{-1}(U) \cap A$ and $U_B := f^{-1}(U) \cap B$. Since both $f|_A$ and $f|_B$ are continuous we find that $U_A = (f|_A)^{-1}(U)$ is open in A and $U_B = (f|_A)^{-1}(U)$ is open in B. Since $A \mid B$ is a partition of X both U_A and U_B are also open in X. Therefore, $f^{-1}(U) = U_A \cup U_B$ is open in X. ■

Exercise 3.

Suppose that for each $i \in \{1, \dots, n\}$ that A_i is a connected subspace of a metric space X, such that $A_i \cap A_{i+1} \neq \varnothing$ for each $i \in \{1, \dots, n-1\}$. Prove that $\bigcup_{i=1}^n A_i$ is connected. Does this result extend to an infinite sequence (A_i) of connected subsets?

Solution. Suppose $f\colon \bigcup_{i=1}^n A_i \to \{0,1\}$ is continuous. The restriction $f|_{A_i} : A_i \to \{0,1\}$ is continuous and therefore constant since A_i is connected. Let us write c_i for the value $f|_{A_i}(x)$ (for any $x \in A_i$). Since the intersection $A_i \cap A_{i+1}$ is not empty, then $c_i = c_{i+1}$ for all i. We conclude that all the c_i are equal and thus that f is constant.

The infinite case is also true, with the same proof. ∎

Exercise 4.

Show that a metric space X is connected if and only if for every $W \subseteq X$ such that $W \neq \varnothing, X$, the boundary ∂W is non-empty.

Solution. A metric space X is connected if and only if the only subsets which are both open and closed $\varnothing$ and X. But W is both open and closed if and only if $\overline{W} = W^\circ$, if and only if $\partial W = \varnothing$. ∎

Exercise 5.

If X is a connected metric space containing more than one point, show that X is infinite.

Solution. Since X is a metric space, all singletons $\{y\}$ are closed. If X is finite, then $X \setminus \{x\} = \bigcup_{y \neq x} \{y\}$ is a finite union of closed sets and is therefore closed. Therefore, $\{x\} = X \setminus (X \setminus \{x\})$ is both open and closed and X cannot be connected (unless $X = \{x\}$). ∎

Exercise 6.

Prove that any continuous function $f\colon [a,b] \to [a,b]$ has a fixed point, that is, a point x such that $f(x) = x$.
Hint: Consider the function $g\colon [a,b] \to \mathbf{R}$ given by $g(x) = f(x) - x$.

Solution. The function g in the hint is a continuous function from a closed interval to $\mathbf{R}$ and, therefore, its image is a closed interval. Since $a \leqslant f(x) \leqslant b$, for all $x \in [a,b]$, we have

$$0 \leqslant g(a) = f(a) - a \leqslant b - a \quad \text{and} \quad a - b \leqslant g(b) = f(b) - b \leqslant 0.$$

So if $f(a) \neq a$, then $g(a) > 0$, and if $f(b) \neq b$, then $g(b) < 0$. So by the Intermediate Value Theorem, there exists $x \in (a,b)$ such that $g(x) = 0$, i.e. $f(x) = x$. ∎

Exercise 7.

a. Prove that if X is a connected metric space and there exists a non-constant continuous function $f\colon X \to \mathbf{R}$, then X is uncountable.

b. Prove that a connected metric space containing more than one point contains uncountably many points.

Solution. a. If f takes two real values $a < b$, then by the Intermediate Value Theorem, it takes on all values in $[a, b]$, which is uncountable, and hence so is X.

b. Apply part a to $f\colon X \to \mathbf{R}$ given by $f(x) = d(x, a)$ for a fixed $a \in X$. ∎

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Show that there is no continuous injective function $f\colon \mathbf{R}^2 \to \mathbf{R}$.
Hint: Consider a triangle in $\mathbf{R}^2$.

Solution. Consider a triangle ABC in $\mathbf{R}^2$. Then BC and $BA \cup AC$ are both connected. So both $f(BC)$ and $f(BA \cup AC)$ contain $[f(B), f(C)]$ by the Intermediate Value Theorem. ■

Exercise 2.

a. Suppose that $f\colon X \to Y$ is a continuous function between metric spaces and that X is path-connected. Show that $f(X)$ is path-connected.

b. Suppose that X and Y are homeomorphic metric spaces. Show that X is path-connected if and only if Y is path-connected.

c. Show that the graph G_f of a continuous function with path-connected domain is path-connected.

Solution.

a. Let $y_1, y_2 \in f(X)$. Then there exists $x_1, x_2 \in X$ such that $y_1 = f(x_1)$ and $y_2 = f(x_2)$. Since X is path-connected, there exists a path $g\colon I \to X$ from x_1 to x_2. The composition $f \circ g\colon I \to Y$ is continuous, as a composition of continuous functions, and therefore gives a path in Y from y_1 to y_2.

b. Suppose that $f\colon X \to Y$ is a homeomorphism. By the first part, if X is path-connected then $f(X) = Y$ is path-connected. Similarly, if Y is path-connected then $X = f^{-1}(Y)$ is path-connected.

c. Recall that the graph G_f of a continuous function is homeomorphic to its domain. Therefore, if the domain is path-connected, then G_f is path-connected by the first part. ■

Exercise 3.

Given two compact subspaces A and B of a metric space X, prove that $A \cup B$ and $A \cap B$ are compact.
Give an example showing that infinite union of compact subsets are not necessarily compact.

Solution. Let $\mathcal{U}$ be an open cover of $A \cup B$. Then $\mathcal{U}$ is an open cover of both A and B so there exist finite subcovers $\mathcal{U}_A$ and $\mathcal{U}_B$ of A and B respectively. So $\mathcal{U}_A \cup \mathcal{U}_B$ is a finite subcover of $\mathcal{U}$.

Now $A \cap B$ is closed subspace of $A \cup B$ since A and B are both closed being compact. Therefore $A \cap B$ is a closed subspace of a compact space $A \cup B$, and therefore compact.

The intervals $[-n, n]$ are compact subsets of $\mathbf{R}$, but their union is $\mathbf{R}$ which is unbounded, hence not compact. For an other example, one can take the intervals $[\frac{1}{n}, 1]$ which are all compact but whose union is $(0, 1]$ which is not closed, hence not compact. Finally, the singletons $\{x\}$ are always closed compact, but their union $\bigcup_{x \in X}\{x\} = X$ does not need to be. ■

Exercise 4.

Prove that a discrete metric space is compact if and only if it is finite.

Solution. Suppose X is discrete and compact. Then $\mathcal{U} := \{\{x\} \mid x \in X\}$ is an open cover of X and therefore has a finite subcover which implies that X is finite.

If a discrete space is finite then it is compact as $\{\{x\} \mid x \in X\}$ is a finite subcover of any open cover of X. ■

Exercise 5.

Let (X, d) be a compact metric space and suppose that $f: X \to X$ is a continuous map such that $f(x) \neq x$, for all $x \in X$. Prove that there exists $\varepsilon > 0$ such that $d(f(x), x) \geqslant \varepsilon$, for all $x \in X$.

Hint: Consider $g: X \to \mathbf{R}$, $g(x) = d(f(x), x)$.

Solution. The function $g: X \to \mathbf{R}$ given by $g(x) = d(f(x), x)$ is continuous being a composition of continuous functions ($g = d \circ (f \times \mathrm{Id}) \circ \Delta$ where $\Delta(x) = (x, x)$). Since $f(x) \neq x$, for all $x \in X$, $g(X) \in (0, +\infty)$. Since X is compact, g attains its bounds on X and so in particular there exists $\varepsilon > 0$ such that $\varepsilon = \inf g(X)$ which gives the result. ■

***Exercise 6.**

Suppose $f, g: I \to X$ are paths in a metric space X such that $f(1) = g(0)$. Show that

$$h(x) := \begin{cases} f(2x), & x \in [0, \frac{1}{2}]; \\ g(2x-1), & x \in [\frac{1}{2}, 1], \end{cases}$$

is a path in X from $f(0)$ to $g(1)$.

Solution. The map h is well defined, $h(0) = f(0)$ and $h(1) = g(1)$. We hence need to show that it is continuous.

First, observe that if $f: [a, b] \to X$ is continuous, so are $f_r: [a + r, b + r] \to X$ and $f_\lambda: [\lambda a, \lambda b] \to X$ for any $r \in \mathbf{R}$ and $0 \neq \lambda \in \mathbf{R}$, where $f_r(x) = f(x-r)$ and $f_\lambda(x) = f(\frac{\lambda}{x})$.

Using the above observation, we want to show that if $f: [0, 1] \to \mathbf{R}$ and $g: [1, 2] \to \mathbf{R}$ are continuous, then so is the map $h: [0, 2] \to \mathbf{R}$ defined by $h(x) = f(x)$ if $x \leqslant 1$ and $h(x) = g(x)$ if $x \geqslant 1$. Let U be an open subset of X. Since $h[0, 2] = f[0, 1] \cup g[1, 2]$ we have $h^{-1}(U) = f^{-1}(U) \cup g^{-1}(U)$. The subset $f^{-1}(U)$ is open in $[0, 1]$ by continuity of f. It is thus a union of balls, that is of the form $\bigcup_{i \in I}(B_i \cap [0, 1])$, where the B_i are open intervals of $\mathbf{R}$. This implies that $f^{-1}(U)$ is open in $[0, 2]$ if and only if it does not contains

1. A similar statement is true for $g^{-1}(U) = \bigcup_{j\in J}(C_j \cap [1,2]$ for some open intervals C_j. If none of $f^{-1}(U)$ and $g^{-1}(U)$ contains 1, they are both open in $[0,2]$ and so is their union. If one of $f^{-1}(U)$ or $g^{-1}(U)$ contains 1 they both do and therefore therefore, for every B_i containing 1 there exists C_j (also containing 1) such that $(B_i \cup C_j) \cap [0,2]$ is open; and vice-versa for C_j and B_i. This implies that $f^{-1}(U) \cup g^{-1}(U)$ is open in $[0,2]$. ∎

***Exercise 7.**

a. Let X be a metric space and show that: "$x \sim y$ if there is a path in X from x to y" is an equivalence relation.

b. Let $\pi(X)$ denote the set of equivalence classes (also known as connected components). For any continuous function $f\colon X \to Y$ we define a function

$$\pi(f)\colon \pi(X) \to \pi(Y)$$

by: if $C(x) \in \pi(X)$ is the connected component containing $x \in X$, then

$$\pi(f)(C(x)) := C(f(x)).$$

Show that $\pi(f)$ is a well-defined function.

c. Given continuous functions $f\colon X \to Y$ and $g\colon Y \to Z$ show that the following diagram commutes:

$$\begin{tikzcd} \pi(X) \arrow[rr, "\pi(f)"] \arrow[dr, "\pi(g\circ f)"'] & & \pi(Y) \arrow[dl, "\pi(g)"] \\ & \pi(Z) & \end{tikzcd}$$

d. Show that if $\mathrm{Id}\colon X \to X$ is the identity map of X, then $\pi(\mathrm{Id})\colon \pi(X) \to \pi(X)$ is the identity map of $\pi(X)$.

e. Show that if $f\colon X \to Y$ is a homeomorphism, then $\pi(f)\colon \pi(X) \to \pi(Y)$ is a bijection.

Solution. a. The constant path $f\colon I \to X$, $f(t) = x$ for all $t \in I$, shows that $x \sim x$. If $f\colon I \to X$ is a path from x to y, then $g\colon I \to X$, $g(t) = f(1-t)$, gives a path from y to x; so $x \sim y \iff y \sim x$. Finally, Question 6 shows that $x \sim y \,\&\, y \sim z \implies x \sim z$.

b. Suppose that $y \in C(x)$, i.e. $x \sim y$. We need to show that $C(f(x)) = C(f(y))$. Since $x \sim y$, we know that $C := C(x) = C(y)$, where C is a connected component of X. So $f(x), f(y) \in f(C)$ and $f(C)$ is the continuous image of a path-connected space and is therefore path-connected. Hence $C(f(x)) = C(f(y))$.

c. For any $x \in X$,

$$(\pi(g) \circ \pi(f))(C(x)) = \pi(g)(C(f(x))) = C(g(f(x))) \\ = C((g \circ f)(x)) = \pi(g \circ f)(C(x)).$$

So $\pi(g) \circ \pi(f) = \pi(g \circ f)$.

d. For any $x \in X$, we have $\pi(\mathrm{Id})(C(x)) = C(\mathrm{Id}(x)) = C(x)$, so $\pi(\mathrm{Id})$ is the identity map of $\pi(X)$.

e. By the previous two parts,

$$\pi(f) \circ \pi(f^{-1}) = \pi(f \circ f^{-1}) = \pi(\mathrm{Id}) = \mathrm{Id},$$

and similarly, $\pi(f^{-1}) \circ \pi(f) = \mathrm{Id}$. So $\pi(f^{-1})$ is the inverse to $\pi(f)$, and so $\pi(f)$ is a bijection. ■

***Exercise 8.**

Suppose that $f\colon [a,b] \to \mathbf{R}$ is a function such that $|f^{-1}(y)| = 2$, for every y in the image of f. Prove that f cannot be continuous.

Solution. Let $f\colon [a,b] \to \mathbf{R}$ be a continuous function. Since $[a,b]$ is a closed interval, its image is also a closed interval, let us say $[c,d]$.

Now, suppose that $4 \leqslant |f^{-1}(c)| + |f^{-1}(d)| < +\infty$. This implies two things. Firstly, we have $c \neq d$, as otherwise we would have $|[a,b]| = |f^{-1}([c,c])| < \infty$, which is absurd. Secondly, either $f^{-1}(c)$ has at least two different elements, and at least one of them is not in $\{a,b\}$, or $f^{-1}(d)$ has at least two different elements, and at least one of them is not in $\{a,b\}$. Let us suppose that $f^{-1}(d)$ has at least two elements, at least one of them, x being not in $\{a,b\}$ (the proof for $f^{-1}(c)$ is similar).

Let us suppose for a moment that $\{x,b\} \subseteq f^{-1}(d)$. Let $\delta > 0$ be such that $[x-\delta, x+\delta]$ and $[b-\delta, b]$ are disjoint. Since the image of a closed interval by a continuous function is a closed interval, we have

$$f([x-\delta, x]) = [y_1, d], \quad f([x, x+\delta]) = [y_2, d], \quad f([b-\delta, b]) = [y_3, d]$$

for some y_1, y_2 and y_3. Let $y = \max\{y_1, y_2, y_3\}$. Then the three disjoint intervals $[x-\delta, x)$, $(x, x+\delta]$ and $[b-\delta, b]$ all contain a preimage of y. That is, $|f^{-1}(y)| \geqslant 3$.

The cases $\{x,a\} \subseteq f^{-1}(d)$ is similar and we can also find $y \in [c,d]$ with $|f^{-1}(y)| \geqslant 3$.

If $|f^{-1}(d) \setminus \{a,b\}| \geqslant 2$, an easy adaptation of the above proof show the existence of a $y \in [c,d]$ such that $|f^{-1}(y)| \geqslant 4$. ■

To go further

The above proof can easily been adapted to show that if $f\colon [a,b] \to \mathbf{R}$ is continuous with $|f^{-1}(y)|$ constant for all $y \in f([a,b])$, then either $|f^{-1}(y)| = 1$ or $|f^{-1}(y)|$ is infinite.

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Prove that the following are equivalent:

I. X is compact;

II. if $\{V_\alpha \mid \alpha \in I\}$ is a collection of closed subsets of X such that $\bigcap_{\alpha\in J} V_\alpha \neq \varnothing$ for any finite $J \subseteq I$, then $\bigcap_{\alpha\in I} V_\alpha \neq \varnothing$.

Solution. "$\Rightarrow$" Suppose that X is compact. Let $U_\alpha := X \setminus V_\alpha$. The proof is by contrapositive. Suppose that $\bigcap_{\alpha\in I} V_\alpha = \varnothing$. We want to show that there exists a finite $J \subseteq I$ such that $\bigcap_{\alpha\in J} V_\alpha = \varnothing$. We have $X = X \setminus (\bigcap_{\alpha\in I} V_\alpha) = \bigcup_{\alpha\in I} U_\alpha$, i.e. $\{U_\alpha \mid \alpha \in I\}$ is an open cover of X. Since X is compact there exists a finite subcover $\{U_\alpha \mid \alpha \in J\}$, for some finite subset $J \subseteq I$, i.e.

$$X = \bigcup_{\alpha\in J} U_\alpha \quad \Longrightarrow \quad X \setminus \left(\bigcup_{\alpha\in J} U_\alpha\right) = \bigcap_{\alpha\in J} V_\alpha = \varnothing$$

as desired.

"$\Leftarrow$" Let $\mathcal{U} = \{U_\alpha \mid \alpha \in I\}$ be an open cover of X and $V_\alpha := X \setminus U_\alpha$. Then $X = \bigcup_{\alpha\in I} U_\alpha$ and therefore $X \setminus (\bigcup_{\alpha\in I} U_\alpha) = \bigcap_{\alpha\in I} V_\alpha = \varnothing$. So there exists a finite subset $J \subseteq I$ such that $\bigcap_{\alpha\in J} V_\alpha = \varnothing$, which implies that $X = \bigcup_{\alpha\in J} U_\alpha$. So $\{U_\alpha \mid \alpha \in J\}$ is a finite subcover of X. ■

Exercise 2.

Suppose that X is a compact metric space and $\{V_n \mid n \in \mathbf{N}\}$ is a collection of non-empty closed nested subsets of X, that is, $V_{n+1} \subseteq V_n$ for all $n \in \mathbf{N}$. Then $\bigcap_{n\in\mathbf{N}} V_n \neq \varnothing$.
Find a counterexample with X non-compact.

Solution. Follows from the previous exercise.

Let $X = (0, 1]$ with the metric induced from the standard metric on $\mathbf{R}$. Then the subsets $(0, \frac{1}{n}]$ are non-empty closed subsets of X which are nested, but their intersection is empty. ■

Exercise 3.

Finish the proof of Corollary 3.2.5. More precisely, shows that if $f\colon S \to \mathbf{R}$ is a bounded function (S any set, not necessarily a metric space), then $\sup f(X)$ and $\inf f(X)$ are elements of $\overline{f(X)}$.

Solution. By Analysis 1, we have a converging sequence $f(x_n) \to \sup f(X)$, which implies $\sup f(X) \in \overline{\{f(x_n) \mid n \in \mathbf{N}\}} \subseteq \overline{f(X)}$. The proof for the infimum is similar. ■

Exercise 4.

Which of the following subspaces of $\mathbf{R}$ and $\mathbf{R}^2$ are compact?

a. $[0, 1)$;

b. $[0, \infty)$;

c. $\mathbf{Q} \cap [0, 1]$;

d. $\{(x, y) \in \mathbf{R}^2 \mid x^2 + y^2 = 1\}$;

e. $\{(x, y) \in \mathbf{R}^2 \mid |x| + |y| \leqslant 1\}$;

f. $\{(x, y) \in \mathbf{R}^2 \mid x^2 + y^2 < 1\}$;

g. $\{(x, y) \in \mathbf{R}^2 \mid x \geqslant 1, 0 \leqslant y \leqslant \frac{1}{x}\}$.

Solution. a, c and f are not closed, hence not compact.

b and g are not bounded, hence not compact.

d is the image of the compact set $[0, 1\pi]$ by the continuous function $f(x) = (\cos x, \sin x)$, hence compact.

e is the closed ball of radius 1 for the taxicab metric, and hence closed for the Euclidean metric (by topological equivalence). It is contained in $\overline{B}_1(0; d_2)$ hence bounded. It is thus compact by the Heine–Borel theorem. ■

Exercise 5.

In the proof of the Heine-Borel Theorem we used the following fact: if $A \subseteq \mathbf{R}^n$ is bounded, then there exists $a, b \in \mathbf{R}$ such that $A \subseteq [a, b]^n$. Prove this.

Solution. If A is bounded, then there exists $r > 0$ such that $A \subseteq \overline{B}_r(x)$, for some $x \in A$. Taking $a = \min\{x_i - r \mid i = 1, 2, \ldots, n\}$ and $b = \max\{x_i + r \mid i = 1, 2, \ldots, n\}$ we see that $\overline{B}_r(x) \subseteq [a, b]^n$. ■

***Exercise 6.**

Suppose that X is a compact metric space and suppose that $f\colon X \to X$ is a continuous map. Let $X_0 = X$, $X_1 = f(X_0)$ and inductively define $X_{n+1} = f(X_n)$ for $n \geqslant 1$.

a. Show that $A = \bigcap_{n \in \mathbf{N}} X_n$ is non-empty.

   Hint: Remember Question 2.

b. Show further that $f(A) = A$.

   Hint: To show that $a \in A$ is in $f(A)$ apply Question 2 to the sets $V_n = f^{-1}(a) \cap X_n$.

Solution. a. We show that the X_n form a nested sequence of closed subsets of X. Note that $X_1 = f(X_0) = f(X) \subseteq X_0$. Suppose inductively that $X_n \subseteq X_{n-1}$ for some integer $n \geqslant 1$. Then $f(X_n) \subseteq f(X_{n-1})$ which says $X_{n+1} \subseteq X_n$. So by induction $X_n \subseteq X_{n-1}$ for all $n \in \mathbf{N}$. Also, since X is compact and f is continuous, $X_1 = f(X)$ is compact. Suppose that X_n is compact for some integer $n \geqslant 0$. Then since f is continuous, $X_{n+1} = f(X_n)$ is also compact. By induction X_n is compact for all integers $n \geqslant 0$. So each X_n is closed in X. Also, each X_n is non-empty by inductive construction. Now by Question 2, $A = \bigcap_{n\in\mathbf{N}} X_n$ is non-empty.

b. The inclusion $f(A) \subseteq A$ is straightforward: if $a \in A$ then $a \in X_n$ for any integer $n \geqslant 0$, so $f(a) \in f(X_n) = X_{n+1}$ for any integer $n \geqslant 0$. But since $X_0 \supseteq X_1 \supseteq \cdots \supseteq X_n \supseteq \cdots$, we have $\bigcap_{n=0}^{\infty} X_{n+1} = \bigcap_{n=1}^{\infty} X_n = \bigcap_{n=0}^{\infty} X_n = A$, and we see that $f(a) \in A$. To prove the other inclusion we follow the hint: for any $a \in A$ let $V_n = f^{-1}(a) \cap X_n$. Since the X_n are nested so are the V_n. Then $\{a\}$ is closed in A, hence since f is continuous, $f^{-1}(a)$ is closed in X. Also each X_n is closed in X as in a. So each V_n is closed in X. Moreover, for each integer $n \geqslant 0$ we know $a \in X_{n+1} = f(X_n)$ so there exists $x \in X_n$ such that $f(x) = a$. This says that $V_n = f^{-1}(a) \cap X_n \neq \varnothing$. Now by Question 2, $\bigcap_{n=0}^{\infty} V_n$ is non-empty. Let b be a point in this set. Then $b \in V_n = f^{-1}(a) \cap X_n$ for any integer $n \geqslant 0$. Also, $b = f^{-1}(a)$ says that $f(b) = a$, and $b \in X_n$ for all integers $n \geqslant 0$ says that $b \in \bigcap_{n=0}^{\infty} X_n = A$. So $A \subseteq f(A)$. We have now proved that $f(A) = A$. ■

***Exercise 7.**

With the notation of Question 2 (so X is compact), suppose that U is open in X and that $\bigcap_{n\in\mathbf{N}} V_n \subseteq U$. Prove that $V_n \subseteq U$, for some $n \in \mathbf{N}$.
Hint: Consider the sequence of sets $W_n = V_n \cap (X \setminus U)$.

Solution. The W_n are closed in X and are also nested. Therefore, $\bigcap_{n\in\mathbf{N}} W_n \neq \varnothing$ by Question 2, if the W_n are all non-empty. But $\bigcap_{n\in\mathbf{N}} W_n = (\bigcap_{n\in\mathbf{N}} V_n) \cap (X \setminus U)$ which implies that $\bigcap_{n\in\mathbf{N}} V_n \nsubseteq U$ contradicting our assumption. Therefore, at least one of the W_n must be empty implying that $V_n \subseteq U$. ■

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Suppose that $f\colon (0,1) \to \mathbf{R}$ is continuous, monotonic and bounded on $(0,1)$. Prove that f is uniformly continuous.

Solution. Suppose that f is increasing. Then $\inf_{x\in(0,1)} f(x)$ exists by boundedness while $\lim_{x\to 0^+} f(x)$ exists by continuity. If f is increasing they are equal. So we have

$$\lim_{x\to 0^+} f(x) = \inf_{x\in(0,1)} f(x) \quad \text{and} \quad \lim_{x\to 1^-} f(x) = \sup_{x\in(0,1)} f(x).$$

We can now define $\bar{f}\colon [0,1] \to \mathbf{R}$ by

$$\bar{f}(x) := \begin{cases} \inf f(x), & x = 0; \\ f(x), & x \in (0,1); \\ \sup f(x), & x = 1. \end{cases}$$

It follows from the above that $\bar{f}$ is a well-defined continuous function. Since it has a compact domain, it is uniformly continuous. Thus $f = \bar{f}\big|_{(0,1)}$ is uniformly continuous. ■

Exercise 2.

Show that two Lipschitz equivalent metrics are uniformly equivalent.

Solution. Suppose that d_1 and d_2 are Lipschitz equivalent metrics on X, i.e. there exists $h, k \in (0,\infty)$ such that

$$h d_2(x,y) \leqslant d_1(x,y) \leqslant k d_2(x,y), \quad \forall x,y \in X.$$

We want to show that they are uniformly equivalent: for every $\varepsilon > 0$, there exists $\delta_1, \delta_2 > 0$ such that

$$d_1(x,y) < \delta_1 \implies d_2(x,y) < \varepsilon \quad \text{and} \quad d_2(x,y) < \delta_2 \implies d_1(x,y) < \varepsilon.$$

We can take $\delta_1 = h\varepsilon$ and $\delta_2 = \frac{\varepsilon}{k}$. ■

Exercise 3.

Show that for any metric space (X,d), the metrics d and $d'(x,y) := \frac{d(x,y)}{1+d(x,y)}$ are uniformly equivalent but may not be Lipschitz equivalent.

Solution. We need to show that for all $\varepsilon > 0$, there exists $\delta, \delta' > 0$ such that

$$d(x,y) < \delta \implies d'(x,y) < \varepsilon \quad \text{and} \quad d'(x,y) < \delta' \implies d(x,y) < \varepsilon.$$

We can take $\delta = \varepsilon$ and $\delta' = \frac{\varepsilon}{(\varepsilon+1)}$ since

$$d'(x,y) \leqslant d(x,y) < \delta \quad \text{and} \quad d(x,y) = \frac{d'(x,y)}{1-d'(x,y)} < \frac{\delta'}{1-\delta'}.$$

Note that $0 \leqslant d'(x,y) < d(x,y)$, for all $x, y \in X$. On $\mathbf{R}$, if $d = d_{\mathbf{R}}$ and d' were Lipschitz equivalent then there would exist $k > 0$ such that

$$d(x,y) = |x-y| \leqslant kd'(x,y) < k, \quad \forall x, y \in \mathbf{R},$$

which is obviously false. ■

Exercise 4.

Show that the metric on $(0,1)$ given by $d(x,y) := |x^{-1} - y^{-1}|$ is topologically equivalent but not uniformly equivalent to the Euclidean metric on $(0,1)$.

Solution. Let d_2 denote the Euclidean metric on $(0,1)$. Since $d_2(x,y) = |x-y| \leqslant d(x,y)$, if a subset $U \subseteq (0,1)$ is d_2-open, then it is d-open. In other words, the identity map is (d, d_2)-continuous (it is even (d, d_2)-uniformly continuous).

Given $a \in (0,1)$ and $\varepsilon > 0$, let $\delta = \min\{\frac{1}{2}a^2\varepsilon, \frac{a}{2}\}$. If $|x-a| < \delta$, then

$$d(x,a) = \frac{|x-a|}{|xa|} < \frac{2|x-a|}{a^2} < \varepsilon.$$

Therefore, the identity map is (d_2, d)-continuous, i.e. if U is d-open, then it is d_2-open. So d and d_2 are topologically equivalent.

But the identity map is not uniformly (d_2, d)-continuous, since taking $\varepsilon = 1$ it is not possible to find a $\delta > 0$ such that $|x-y| < \delta$ implies $d(x,y) < 1$. ■

Exercise 5.

Prove that if $A \subseteq \mathbf{R}$ is not closed, then there exists a continuous function $f \colon A \to \mathbf{R}$ that is not uniformly continuous.

Solution. If A is not closed then there exists a point $x \in \overline{A} \setminus A$. Define $f \colon A \to \mathbf{R}$ by $f(x) = \frac{1}{x-a}$. Then this is a continuous function that is not uniformly continuous. ■

Exercise 6.

Prove that any Cauchy sequence in a metric space is bounded.

Solution. Let (x_n) be a Cauchy sequence in a metric space (X,d) and take $\varepsilon = 1$. Then there exists $N \in \mathbf{N}$ such that $m, n \geqslant N \implies d(x_m, x_n) < 1$. Let

$$M := \max\{d(x_i, x_N) \mid i \in \{1, \dots, N-1\}\}.$$

Then $d(x_n, x_N) \leqslant \max\{1, M\}$, for all $n \in \mathbf{N}$, and so (x_n) is bounded. ■

Exercise 7.

Give an example of a metric space that is totally bounded but not compact.

Solution. The subspace $(0,1) \subseteq \mathbf{R}$ is not complete, hence not compact. It is totally bounded (in general, bounded and totally bounded are the same in Euclidean space). Indeed, for every $\varepsilon > 0$ there exists N such that $\frac{1}{N} < \varepsilon$. Then $\{\frac{i}{N} \mid 1 \leqslant i \leqslant N-1\}$ is a finite $\frac{1}{N}$-net for $(0,1)$, hence a finite ε-net. ■

Exercise 8.

Find appropriate sequences that prove the following subspaces of Euclidean space are not sequentially compact:

a. $(-6, 9) \subseteq \mathbf{R}$;

b. $\mathbf{Z}^2 \subseteq \mathbf{R}^2$;

c. $\{x \in \mathbf{R}^3 \mid |x_1| > 2\} \subseteq \mathbf{R}^3$;

d. $\{x \in \mathbf{R}^4 \mid x_1x_2x_3x_4 < 0\} \subseteq \mathbf{R}^4$.

Solution. In each subspace we need to find an infinite sequence with no convergent subsequences (in that subspace!) The following are suitable examples (amongst others):

a. $(-6 + \frac{1}{n})$: every subsequence converges in $\mathbf{R}$ to $-6 \notin (-6, 9)$;

b. (n, n): any subsequence is unbounded, thus not Cauchy by Question 6;

c. $(3, 0, n)$: idem;

d. $(n, -1, 1, 1)$: idem. ■

***Exercise 9.**

Suppose that X is a sequentially compact metric space and that $\mathcal{U} = \{U_1, U_2, \dots, U_n\}$ is an open cover of X. Put $C_i := X \setminus U_i$. Show that if $U_i = X$, for some $i \in \{1, \dots, n\}$, then any $\varepsilon > 0$ is a Lebesgue number for $\mathcal{U}$. From now on, assume that no U_i is X.

a. Defining $d(x, C_i) = \inf\{d(x, c) \mid c \in C_i\}$, prove that $x \mapsto d(x, C_i)$ is a continuous real-valued function on X all of whose values arc non-negative.

b. Defining $f(x) = \frac{1}{n}\sum_{i=1}^{n} d(x, C_i)$, show that $f\colon X \to \mathbf{R}$ is continuous with a positive value for each $x \in X$.

c. Deduce that there exists $\varepsilon > 0$ such that $f(x) \geqslant \varepsilon$, for all $x \in X$.

d. For any $x \in X$, show that $f(x) \leqslant \max\{d(x, C_i) \mid i\{1, \dots, n\}\}$.

e. For any $x \in X$, show that $B_\varepsilon(x) \subseteq U_{k(x)}$, where $k(x) \in \{1, \dots, n\}$ is such that $d(x, C_{k(x)}) = \max\{d(x, C_i) \,|\, i \in \{1, \dots, n\}\}$. Deduce that ε is a Lebesgue number for $\mathcal{U}$.

Solution. We can just take $U(x) = U_i$, for all $x \in X$, to show that any $\varepsilon > 0$ is a Lebesgue number if $U_i = X$.

a. Since C_i is closed in X, it is compact, and for every $x \in X$, there is a $c_x \in C_i$ such that $d(x, C_i) = d(x, c_x)$. Therefore, $d(x, c_x) \leqslant d(x, c)$, for every $c \in C_i$. Now let $x \in X$ and $\varepsilon > 0$. Take $\delta = \varepsilon$ and suppose $d(x, y) < \delta$. Suppose, without loss of generality, that $d(x, C_i) \geqslant d(y, C_i)$. Then

$$|d(x, C_i) - d(y, C_i)| = d(x, c_x) - d(y, c_y) \leqslant d(x, c_y) - d(y, c_y) \leqslant d(x, y) < \delta = \varepsilon.$$

So the function is continuous. Since $d(x, C_i)$ is the infimum of a set of non-negative numbers it is also non-negative.

b. By the composition of continuous functions, f is continuous. For $x \in X$, x cannot be in every C_i since otherwise x would not be in any U_i and $\mathcal{U}$ would not be an open cover for X. Therefore, there exists $k \in \{1, \dots, n\}$ such that $x \notin C_k$ and $d(x, C_k) > 0$. Therefore, $f(x) > 0$, for all $x \in X$.

c. Since f is a continuous real-valued function with compact domain it attains its bounds, in particular, its infimum. As it is positive everywhere, this implies that there exists $\varepsilon > 0$ such that $f(x) \geqslant \varepsilon$ for all $x \in X$.

d. We have,

$$\begin{aligned} f(x) = \frac{1}{n} \sum_{i=1}^{n} d(x, C_i) &\leqslant \frac{1}{n} \left(n \cdot \max\{d(x, C_i) \mid i \in \{1, \dots, n\}\}\right) \\ &= \max\{d(x, C_i) \mid i \in \{1, \dots, n\}\}. \end{aligned}$$

e. Suppose that $y \in B_\varepsilon(x)$, that is: $d(x, y) < \varepsilon$. We need to show that $y \in U_{k(x)} = X \setminus C_{k(x)}$. We know that $0 < \varepsilon \leqslant f(x) \leqslant d(x, C_{k(x)})$. If $y \in C_{k(x)}$, then $d(x, C_{k(x)}) < d(x, y) < \varepsilon$ providing a contradiction. ∎

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Show that in any metric space

a. The union of two complete subspaces is complete.

b. The intersection of two complete subspaces is complete.

Hint: Use Lemma 4.1.7 for part a.

Solution. a. Let X and Y be complete metric spaces and (a_n) be a Cauchy sequence in $X \cup Y$. Then (a_n) has infinitely many terms in either X or Y (as an infinite sequence). Suppose (a_n) has infinitely many terms in X, then these terms can be considered as a subsequence of (a_n) in X which will converge to a point in X by completeness. Therefore, (a_n) converges in $X \cup Y$ by Lemma 4.1.7.

b. The intersection $X \cap Y$ is a closed subspace of the complete space X, and is therefore complete. ■

Exercise 2.

Prove that any discrete metric space is complete.

Solution. Let (x_n) be a Cauchy sequence in a discrete metric space X. Taking $\varepsilon = 1$ in the definition of a Cauchy sequence proves that (x_n) is eventually constant and therefore converges. ■

Exercise 3.

Which of the following are complete:

a. $\left\{\frac{1}{n} \,\middle|\, n \in \mathbf{N}\right\} \cup \{0\}$;

b. $\mathbf{Q} \cap [0, 1]$;

c. $\left\{(x, y) \in \mathbf{R}^2 \,\middle|\, x > 0, y \geqslant \frac{1}{x}\right\}$.

Solution. a. Complete (as a closed subspace of $\mathbf{R}$);

b. Not complete (take x_n to be the decimal development up to the n^{th} place of $\frac{\sqrt{2}}{2}$);

c. Complete (as a closed subspace of $\mathbf{R}^2$). ■

Exercise 4.

Show that the uniformly continuous image of a complete metric space is not necessarily complete.

Solution. Define $f\colon [1,+\infty) \to (0,1]$ by $f(x) = \frac{1}{x}$. The space $[1,+\infty)$ is complete as a closed subset of $\mathbf{R}$, while $(0,1]$ is not. One easily check that f is a bijection. It remains to show that f is uniformly continuous. For any $\varepsilon > 0$, for $\delta = \varepsilon$ if $|x-y| < \varepsilon$ then $|f(x) - f(y)| = \frac{1}{|xy|}|x-y| < \frac{1}{|xy|}\varepsilon \leqslant \varepsilon$. ∎

Exercise 5.

Give an example of topologically equivalent metrics d and d' on a set X and a sequence (x_n) in X that is Cauchy in (X,d) but not in (X,d').
Hint: See Tutorial 9, Question 4 for an example of topologically equivalent metrics that are not uniformly equivalent.

Solution. On $(0,1)$, take the standard metric d_2 and $d(x,y) = |x^{-1} - y^{-1}|$. These two metrics are topologically equivalent by Tutorial 9, Question 4. Then the sequence $(\frac{1}{n})$ is d_2-Cauchy but is not d-Cauchy. Indeed, $d(\frac{1}{n}, \frac{1}{m}) = |m-n|$. So for $\varepsilon = 1$, for every N there exists $n = N$ and $m = N+2$ bigger than N such that $d(n,m) = 2 > \varepsilon$. ∎

***Exercise 6.**

Which of the following metrics on $\mathbf{R}$ give complete metric spaces?

a. $d(x,y) = |x^3 - y^3|$;

b. $d(x,y) = |\mathrm{e}^x - \mathrm{e}^y|$;

c. $d(x,y) = |\tan^{-1}(x) - \tan^{-1}(y)|$.

Solution. Let us label these metrics d_a, d_b, d_c.

a. Then $(\mathbf{R}, d_a)$ is complete. For let (x_n) be a d_a-Cauchy sequence. Then (x_n^3) is a Cauchy sequence in the ordinary sense, so it converges say to $y \in \mathbf{R}$. Let $x = y^{\frac{1}{3}}$. Then (x_n) converges to x in the metric d_a. This holds since
$$d_a(x_n, x) = |x_n^3 - x^3| = |x_n^3 - y| \to 0, \quad \text{as } n \to \infty.$$

b. This is not complete. Consider the sequence $(-n)$. Then $\mathrm{e}^{-n} \to 0$ as $n \to \infty$, so (e^{-n}) is Cauchy in the usual sense, so $(-n)$ is d_b-Cauchy. But suppose that $(-n)$ converged to a real number x_0 in $(\mathbf{R}, d_b)$. Then $d_b(-n, x_0) \to 0$ as $n \to \infty$, which says that (e^{-n}) converges to e^{x_0} in the usual metric on $\mathbf{R}$. But we know that (e^{-n}) converges to 0 in the usual metric, so would have $\mathrm{e}^{x_0} = 0$. But there is no such real number x_0, so $(-n)$ cannot converge in $(\mathbf{R}, d_b)$. Hence $(\mathbf{R}, d_b)$ is not complete.

c. For similar reasons this is not complete either. Consider the sequence (n) in $\mathbf{R}$. The sequence $(\tan^{-1}(n))$ converges to $\frac{\pi}{2}$ in the usual sense as $n \to \infty$. So $(\tan^{-1}(n))$ is Cauchy in the usual sense which says that (n) is d_c-Cauchy. But suppose that (n) converges to some $x_0 \in \mathbf{R}$ in the metric space $(\mathbf{R}, d_c)$. Then $(\tan^{-1}(n))$ converges to $\tan^{-1}(x_0)$ in the usual sense. This gives $\tan^{-1}(x_0) = \frac{\pi}{2}$. But there is no such real number x_0. So (n) does not converge in $(\mathbf{R}, d_c)$ and this space is not complete.

■

***Exercise 7.**

Suppose that $f\colon X \to Y$ is a bijection between metric spaces such that f is uniformly continuous and f^{-1} is continuous.

a. Prove that if (x_n) is a Cauchy sequence in X, then $f(x_n)$ is a Cauchy sequence in Y.

b. Prove that if y_n is a convergent sequence in Y then $(f^{-1}(y_n))$ is a convergent sequence in X.

c. Deduce that if Y is complete, then X is complete.

d. Deduce that completeness is an invariant of uniform equivalence.

Solution. a. Let $\varepsilon > 0$. Then, by the uniform continuity of f, there is a $\delta > 0$ such that $d_X(x,y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon$. For this δ, there is an $N \in \mathbf{N}$, such that $m, n \geqslant N \implies d_X(x_m, x_n) < \delta$ since (x_n) is Cauchy. Therefore, $m, n \geqslant N \implies d_Y(f(x_m), f(x_n)) < \varepsilon$.

b. Follows from the (sequential) continuity of f^{-1}.

c. Let (x_n) be a Cauchy sequence in X. Then $f(x_n)$ is a Cauchy sequence in Y by a which implies that $f(x_n)$ converges in Y since we are assuming Y is complete. Therefore, by b, $(f^{-1}(f(x_n))) = (x_n)$ converges in X and so X is complete.

d. If we suppose that f is now a uniform equivalence we immediately obtain that if Y is complete, then X is complete. By considering f^{-1} we also get the converse. ■

Exercise 1.

Let $f\colon (0,\frac{1}{4}) \to (0,\frac{1}{4})$ be given by $f(x) = x^2$. Show that f is a contraction without a fixed point.

Solution. Firstly, $f(0,\frac{1}{4}) \subseteq (0,\frac{1}{4})$ so f is a self-map. Also, f is a contraction as

$$|f(x) - f(y)| = |x^2 - y^2| = (x+y)\cdot|x-y| < \frac{1}{2}|x-y|.$$

It has no fixed point since $x^2 = x \implies x \in \{0,1\}$ but neither 0 nor 1 is in $(0,\frac{1}{4})$. ■

Exercise 2.

Let $X = \{x,y,z\}$ and let d be the discrete metric on X. Define

$$d'(x,y) = 2 = d'(x,z), \quad d'(y,z) = 1, \quad \text{and} \quad d'(x,x) = d'(y,y) = d'(z,z) = 0.$$

Show that d' is a metric on X that is Lipschitz equivalent to d. Define $f\colon X \to X$ by

$$f(x) = y, \quad f(y) = z \quad \text{and} \quad f(z) = z.$$

Show that f is a d'-contraction but not a d-contraction.

Solution. It is easy to show that d' is a metric on X. To show that it is Lipschitz equivalent to d note that

$$\frac{1}{2}d'(a,b) \leqslant d(a,b) \leqslant d'(a,b) \quad \forall a,b \in X.$$

By putting $K = \frac{1}{2}$ we can see that f is a d'-contraction by checking that $d(f(a),f(b)) \leqslant \frac{1}{2}d(a,b)$ for any elements $a,b \in X$. Since $d(f(a),f(a)) = 0$ for all a and since d is symmetric, we in fact only need to check this inequality for the three pairs (x,y), (x,z) and (y,z).

To show that f is not a d-contraction observe that

$$d(f(x),f(y)) = d(y,z) = 1 \leqslant Kd(x,y) = K \implies K \geqslant 1.$$

Since K has to be less than 1, f is not a d-contraction. ■

Exercise 3.

Let $f\colon X \to X$ be a map of a compact metric space X such that

$$d(f(x),f(y)) < d(x,y)$$

for any distinct points $x,y \in X$. Prove that f has a unique fixed point.
Hint: Show that $\inf\{d(x,f(x)) \mid x \in X\}$ *is attained, and get a contradiction unless this* inf *is zero.*

Solution. Consider the composition

$$d \circ (1 \times f) \circ \Delta \colon X \longrightarrow X \times X \longrightarrow X \times X \longrightarrow \mathbf{R}$$
$$x \longmapsto (x, x) \longmapsto (x, f(x)) \longmapsto d(x, f(x))$$

Each map here is continuous so the composition, F say, is continuous. Since X is compact, F attains its lower bound l, say, on X. let $p \in X$ be an element such that $F(p) = l$. Suppose by contradiction that $l > 0$, so $d(p, f(p)) = l$. Then p and $f(p)$ are distinct, so by assumption $l \leqslant F(f(p)) = d(f(p), f(f(p))) < d(p, f(p)) = l$. This contradiction shows that l must be 0 and so $f(p) = p$ and we have proved the existence of a fixed point.

Uniqueness: for distinct fixed points $p \neq q$ we would have

$$d(p, q) = f(f(p), f(q)) < d(p, q),$$

a contradiction. So p is the unique fixed point. ■

Exercise 4.

Suppose that $A \subseteq \mathbf{R}$ is of the form $A = \bigcup_{i=1}^{r} A_i$ and let $(f_n \colon A \to \mathbf{R})$ be a sequence of functions. Suppose that for each $i \in \{1, \dots, r\}$, the sequence $(f_n|_{A_i} \colon A_i \to \mathbf{R})$ is uniformly convergent on A_i. Show that (f_n) is uniformly convergent on A.

Solution. Let g_i be the uniform limit of $f_n|_{A_i}$. We claim that if $x \in A_i \cap A_j$, then $g_i(x) = g_j(x)$. Indeed, g_i being the uniform limit of the $f_n|_{A_i}$ it is also their pointwise limit. So for $x \in A_I$ we have $g_i(x) = \lim_{n\to\infty} f_n|_{A_i}(x) = \lim_{n\to\infty} f_n(x)$. This shows that the function $g(x) = g_i(x)$ if $x \in A_i$ is well-defined and is the pointwise limit of (f_n). We will show it is also the uniform limit.

Let $\varepsilon > 0$. For each $i \in \{1, \dots, r\}$ the sequence (f_n) converges uniformly on A_i, so there exists $N_i \in \mathbf{N}$ such that $|f_n(x) - g(x)| < \varepsilon$ whenever $n \geqslant N_i$ and $x \in A_i$. Let $N = \max\{N_1, N_2, \dots, N_r\}$, let x be any point in A and let $n \geqslant N$. Then $x \in A_i$ for some $i \in \{1, \dots, r\}$ and $n \geqslant N \geqslant N_i$, so $|f_n(x) - g(x)| < \varepsilon$. Thus (f_n) converges uniformly on A. ■

Exercise 5.

Let $A \subseteq \mathbf{R}$ and let $f_n \colon A \to \mathbf{R}$ be a sequence of bounded functions. Show that

a. If (f_n) uniformly converges to $f \colon A \to \mathbf{R}$, then f is bounded;

b. If (f_n) is uniformly Cauchy on A, then it is uniformly bounded. Meaning that there exists $M \in \mathbf{R}$ such that

$$|f_n(x)| \leqslant M, \quad \text{for all } x \in A \text{ and all } n \in \mathbf{N}.$$

Solution. a. There exists $N \in \mathbf{N}$ such that $n \geqslant N \implies |f_n(x) - f(x)| < 1$, for all $x \in A$. Let $|f_N(x)| \leqslant M_N$, for all $x \in A$. Then

$$|f(x)| < 1 + |f_N(x)| \leqslant 1 + M_N, \quad \forall x \in A.$$

b. Using the same idea as in part a, we see that for $\varepsilon = 1$ there exists N such that for $m = N$

$$n \geqslant N \implies |f_n(x)| < 1 + |f_N(x)| \leqslant 1 + M_N, \quad \forall x \in A.$$

For each $n < N$, we know that there exists M_n such that $|f_n(x)| \leqslant M_n$, for all $x \in A$. So we can take $M := \max\{1 + M_N, M_n \mid n \in \{1, \dots, N-1\}\}$. ■

Exercise 6.

Which of the following sequences (f_n) converge uniformly on $[0, 1]$?

a. $f_n(x) = \frac{x}{1+nx}$;

b. $f_n(x) = nx\mathrm{e}^{-nx^2}$;

c. $f_n(x) = n^{\frac{1}{2}}x(1-x)^n$;

d. $f_n(x) = nx(1-x^2)^{n^2}$;

e. $f_n(x) = \frac{x^n}{1+x^n}$;

f. $f_n(x) = n^{-x}x^n \cos(nx)$.

Solution. The pointwise limits are the zero functions except for e where the pointwise limit is $f(x) = 0, x \in [0, 1), f(1) = \frac{1}{2}$, which is discontinuous and hence the convergence is not uniform. Also non-uniform are b and d. Everything else converges uniformly. For f, use $|n^{-x}x^n \cos(nx)| \leqslant \frac{1}{2}^n$ on $[0, \frac{1}{2}]$, and $|n^{-x}x^n \cos(nx)| \leqslant n^{-\frac{1}{2}}$ on $[\frac{1}{2}, 1]$.

Below are the details. We will use extensively that for a sequence of functions $(f_n : A \to \mathbf{R})$ pointwise converging to the 0 function on A, the convergence is uniform if and only if $M_n := \sup\{|f_n(x)| \mid x \in A\}$ converges to 0. Let us justify this claim. On one hand, suppose that M_n does converge to 0. So for every $\varepsilon > 0$ there exists N such that $n \geqslant N$ implies $M_n < \varepsilon$. But then, for every $x \in A$ we have $|f_n(x)| \leqslant M_n < \varepsilon$, which shows the uniform convergences of the f_n to 0. On the other hand, let us suppose that we have uniform convergence to 0. So for every $\varepsilon > 0$ there exists N such that $n \geqslant N$ implies $|f_n(x)| < \frac{1}{2}\varepsilon$ for all x. This inequality being true for all x we have $M_n \leqslant \frac{1}{2}\varepsilon < \varepsilon$ and thus the M_n converges to 0.

a. Since $0 \leqslant \frac{x}{1+nx} \leqslant \frac{1}{n}$ for all $x \in [0, 1]$ (including $x = 0$) it follows that the pointwise limit of (f_n) here is the zero function. The same inequality shows that the maximum M_n of f_n on $[0, 1]$ satisfies $M_n \leqslant \frac{1}{n}$ so $M_n \to 0$ as $n \to \infty$ and convergence is uniform on $[0, 1]$.

b. Again the pointwise limit exists and is the zero function: for $f_n(0) = 0$ for all $n \in \mathbf{N}$, and when $x \in (0,1]$ we have $0 < f_n(x) < \frac{2nx}{(nx^2)^2} = \frac{2}{nx^3}$ (using the first terms in the series expansion of e). But $f_n'(x) = n\mathrm{e}^{-nx^2} - 2n^2x^2\mathrm{e}^{-nx^2}$ which is zero when, and only when, $x = \sqrt{\frac{1}{2n}}$. Also, $f_n'(x) > 0$ for $x < \sqrt{\frac{1}{2n}}$ and $f_n'(x) < 0$ for $x = \sqrt{\frac{1}{2n}}$, so the maximum M_n of f_n on $[0,1]$ is attained at $\sqrt{\frac{1}{2n}}$. Hence $M_n = \sqrt{\frac{n}{2}}\mathrm{e}^{-\frac{1}{2}}$. Since M_n does not converge to 0 as $n \to \infty$ convergence is not uniform on $[0,1]$.

c. Again $f_n(0) = f_n(1) = 0$ for all $n \in \mathbf{N}$ and $f_n(x) \to 0$ as $n \to \infty$ for $x \in (0,1)$, so again the pointwise limit is the zero function. Now $f_n'(x) = n^{\frac{1}{2}}(1-x)^n - n^{\frac{3}{2}}x(1-x)^{n-1}$. This has a unique zero: $x = \frac{1}{n+1}$. Since $f_n(\frac{1}{n+1}) > 0$ and $f_n(0) = f_n(1) = 0$, this is the maximum and we have

$$M_n = f_n\left(\frac{1}{n+1}\right) = \frac{n^{\frac{1}{2}}}{n+1}\left(\frac{n}{n+1}\right)^n \to 0$$

as $n \to \infty$. So convergence is uniform on $[0,1]$.

d. Again as before (f_n) converges pointwise to the zero function. The derivative is $f_n'(x) = n(1-x^2)^{n^2} - 2x^2n^3(1-x^2)^{n^2-1}$, which has a unique zero: $x = \frac{1}{\sqrt{2n^2+1}}$. As before we have

$$M_n = f_n\left(\frac{1}{\sqrt{2n^2+1}}\right) = \frac{n}{\sqrt{2n^2+1}}\left(\frac{2n^2}{2n^2+1}\right)^{n^2} \to \frac{1}{\sqrt{2e}}$$

as $n \to \infty$. Hence convergence is not uniform on $[0,1]$ in this case.

e. Here the pointwise limit function is discontinuous, it takes the value 0 when $x \in [0,1)$ but the value $\frac{1}{2}$ when $x = 1$. Each f_n is continuous on $[0,1]$ so convergence is not uniform on $[0,1]$.

f. The pointwise limit is again the zero function (we can argue separately for $x \in [0,1)$ and for $x = 1$). The function f_n satisfies $|f_n(x)| \leqslant x^n \leqslant \frac{1}{2^n}$ for $x \in [0,\frac{1}{2}]$, while $|f_n(x)| \leqslant \frac{1}{n^{\frac{1}{2}}}$ for $x \in [\frac{1}{2},1]$. These give uniform convergence on $[0,\frac{1}{2}]$ and $[\frac{1}{2},1]$ respectively hence we get uniform convergence on $[0,1]$. ■

Exercise 7.

Construct a sequence of functions $(f_n : \mathbf{R} \to \mathbf{R})$ none of which are continuous at $x = 0$, but such that (f_n) converges uniformly on $\mathbf{R}$ to a continuous function.

Solution. Take

$$f_n(x) = \begin{cases} \frac{1}{n}, & x \in \mathbf{Q}; \\ 0, & x \notin \mathbf{Q}. \end{cases}$$ ■

Exercise 1.

Let $y(t) = \mathrm{e}^t$ with $t_0 = 0$. Then calculate the first few Picard iterates $\varphi_n(t)$ of the corresponding IVP

$$\begin{cases} y' = y \\ y(0) = 1 \end{cases}$$

and check that it is converging to the well-known series expansion of e.

Solution. We have $y' = f(y,t)$ for $f(y,t) = y$ and $y_0 = y(t_0) = 1$. So $\varphi_0(t) = 1$ and

$$\varphi_1(t) = 1 + \int_0^t \mathrm{d}s = 1 + t; \quad \varphi_2(t) = 1 + \int_0^t (1+s)\,\mathrm{d}s = 1 + t + \frac{t^2}{2};$$
$$\varphi_3(t) = 1 + \int_0^t (1 + s + \frac{s^2}{2})\,\mathrm{d}s = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}.$$

We know that $\mathrm{e}^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. ■

Exercise 2.

Determine whether or not the real numbers $\frac{8}{81}, \frac{1}{4}, 0.296296296\ldots$, $\frac{\pi}{4}$ and $\frac{\sqrt{3}}{2}$ lie in the Cantor Set $K \subseteq \mathbf{R}$.

Solution. To compute the ternary expansion of $x = \frac{8}{81}$, write

$$x = \frac{(2\times 3)+2}{81} = \frac{0}{3} + \frac{0}{9} + \frac{2}{27} + \frac{2}{81} = 0.002200\ldots\ ;$$

since there are no 1s, it follows that $\frac{8}{81} \in K$.

For the ternary expansion $x = \frac{1}{4} = 0.x_1x_2x_3\ldots$, note that x_j are given by iterating the following procedure: multiply x by 3, record the integer part, take the fractional part, and repeat. So $\frac{3}{4} = x_1 \cdot x_2x_3\ldots$, and so $x_1 = 0$; then $\times 3$ gives $2\frac{1}{4} = \frac{9}{4} = x_2 \cdot x_3x_4\ldots$, so $x_2 = 2$, and $\frac{1}{4} = 0.x_3x_4\ldots$. Repeating forever gives $\frac{1}{4} = 0.0202020\ldots$, which therefore lies in K.

For $x = 0.296296296\ldots$, note that $10^3x = 296 + x$, so $x = \frac{296}{999} = \frac{8}{27}$. The ternary expansion is given by

$$x = \frac{(2\times 3)+2}{27} = \frac{2}{9} + \frac{2}{27} = 0.02200\ldots\ ,$$

so $0.296296296\ldots \in K$.

For $x = \frac{\pi}{4}$ and $\frac{\sqrt{3}}{2}$, follow the same procedure as for $\frac{1}{4}$. So $x = \frac{\pi}{4} = 0.785398\ldots$ implies that x_1 is the integer part of $3x = 2.35619\ldots$, and x_2 is the integer part of $3 \times 0.35619\ldots = 1.0685\ldots$. Since $\frac{\pi}{4}$ is irrational, $x_3x_4\ldots$ cannot give $22\ldots$ as othrwise one would have $\frac{\pi}{4} = 0.22$. Thus any ternary expansion of $\frac{\pi}{4}$ begins by $0.21\ldots$, and thus $\frac{\pi}{4} \notin K$.

Similarly, $x = \frac{\sqrt{3}}{2} = 0.866025\ldots$ implies that x_1 is the integer part of $3x = 2.59807\ldots$, and x_2 is the integer part of $3 \times 0.59807\ldots = 1.7942\ldots$. Since $\frac{\sqrt{3}}{2}$ is irrational, $x_3x_4\ldots$ cannot give $22\ldots$. Thus any ternary expansion begins by $0.21\ldots$, and $\frac{\sqrt{3}}{2} \notin K$. ■

Exercise 3.

The set $\mathrm{Av} \subseteq K$ of *averages* in the Cantor Set is defined by

$$\mathrm{Av} := \left\{ \frac{a+b}{2} \,\middle|\, a, b \in K, a \neq b \right\} \cap K \subseteq \mathbf{R}.$$

Prove that $\mathrm{Av} = \left\{ \frac{k}{3^m} \,\middle|\, 1 \leqslant m, 1 \leqslant k \leqslant 3^m, k \not\equiv 0 \mod 3 \right\} \cap K$.

Solution. Let $a, b \in K$ have ternary expansions $0.a_1a_2\ldots$, $0.b_1b_2\ldots$, where $a \neq b$. Because $a_n, b_n \neq 1$ for any n, it follows that $\frac{a_n+b_n}{2} = 0, 1$ or 2, and therefore that the ternary expansion of $c := \frac{a+b}{2}$ has $c_n = \frac{a_n+b_n}{2}$. Moreover, there exists a least value of m for which $a_m \neq b_m$, and so $c_m = 1$. In other words, $c = c_1 \ldots c_{m-1}c_m \ldots$ with the $c_j \in \{0, 2\}$ for $j < m$. Since $c \in K$, it admits a ternary expansion without 1. This is possible if and only if $c = c_1 \ldots c_{m-1}c_m222\ldots = c_1 \ldots c_{m-1}2000\ldots 0$. But then $c = \frac{k}{3^m}$ for $k = 2 + c_{m-1}3 + \cdots + c_1 3^{m-1}$ not divisible by 3. This shows that $\mathrm{Av} \subseteq \left\{ \frac{k}{3^m} \,\middle|\, 1 \leqslant m, 1 \leqslant k \leqslant 3^m, k \not\equiv 0 \mod 3 \right\} \cap K$

For the other inclusion, let $c = \frac{k}{3^m}$ be in K for some k not divisible by 3. In other words, $c = 0.c_1 \ldots c_{m-1}$ with $c_j \in \{0, 2\}$ and $c_m = 2$. We now construct a and b by:

$$\begin{cases} a_j = b_j = c_j & j < m \\ a_j = 0, b_j = 2 & j = m \\ a_j = b_j = 2 & j > m. \end{cases}$$

By construction, both $a \neq b$ both belong to K. Finally, their average is

$$\frac{0.c_1 \ldots c_{m-1}0222\ldots + 0.c_1 \ldots c_{m-1}2222\ldots}{2} = 0.c_1 \ldots c_{m-1}122\ldots = c.$$

■

Exercise 4.

Let (X, d) be a metric space with a non-empty subset $B \subseteq X$ and a point $x \in X$. Recall that the distance between x and B is defined to be

$$d(x, B) := \inf\{d(x, b) \mid b \in B\}.$$

a. Find an example with $x \notin B$ but $d(x, B) = 0$.

b. Show that if B is compact, then there exists $b^* \in B$ such that $d(x, B) = d(x, b^*)$.

Solution. a. Take $x = 1$ and $B = [0, 1) \in \mathbf{R}$. More generally, any $x \in \overline{B} \setminus B$ will do.

b. By definition of the infimum, for any $\varepsilon > 0$ there exists $b_\varepsilon \in B$ such that $d(x, b_\varepsilon) < d(x, B) + \varepsilon$. We can hence construct a sequence (b_n) with $d(x, b_n) < d(x, B) + \frac{1}{n}$. By compacity, this sequence admits a convergent subsequence with limit b^*. We have $d(x, B) \leqslant d(x, b^*) \leqslant d(x, B) + \frac{1}{n}$ for every n and thus $d(x, B) = d(x, b^*)$. ■

Exercise 5.

Recall that for $A \subseteq X$ and $\varepsilon > 0$, the ε-neighbourhood of A is given by

$$A_\varepsilon := \{x \in X \mid d(x, A) < \varepsilon\}.$$

Show that for non-empty compact subsets of X we have $d_H(A, B) < \varepsilon$ if and only if $A \subseteq B_\varepsilon$ and $B \subseteq A_\varepsilon$.

Solution. We have $d_H(A, B) = \max\{d(A, B), d(B, A)\}$, where $d(A, B) = \sup_{a \in A} d(a, B)$.

If $d(A, B) < \varepsilon$, then $d(a, B) < \varepsilon$ for all $a \in A$ and therefore $A \subseteq B_\varepsilon$. For the converse, if $A \subseteq B_\varepsilon$, then $d(a, B) < \varepsilon$ for all $a \in A$ and therefore $d(A, B) \leqslant \varepsilon$. The problem here is that we would like to have $d(A, B) < \varepsilon$. This is true if A is closed as in this case the supremum is attained.

Similarly, one have $d(B, A) < \varepsilon$ if and only if $B \subseteq A_\varepsilon$. The formula for $d_H(A, B)$ follows. ■

To go further

It is important that A and B are closed. Indeed, for $A = [0, 1)$ and $B = (-1, 0]$ in $\mathbf{R}$ we have $d_H(A, B) = 1$ while $A \subseteq B_1$ and $B \subseteq A_1$.

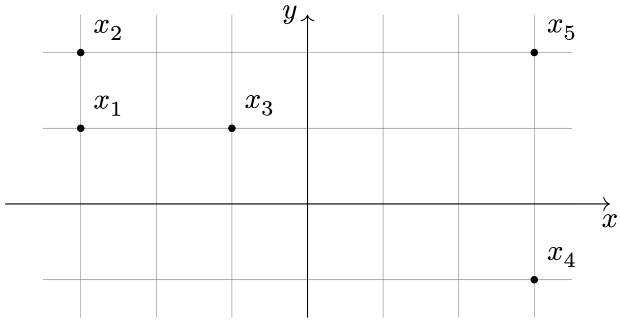
Exercise 6.

Consider the following five points in $(\mathbf{R}^2, d_2)$:

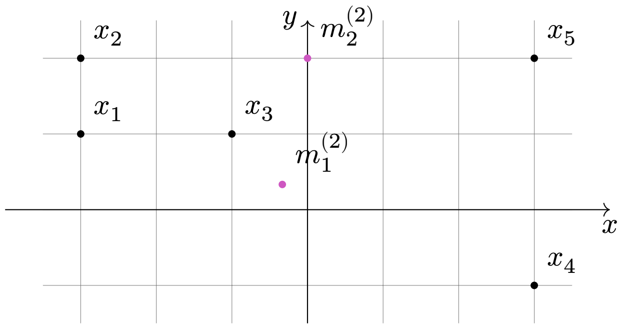
$$x_1 = (-3, 1), \quad x_2 = (-3, 2), \quad x_3 = (-1, 1), \quad x_4 = (3, -1), \quad x_5 = (3, 2).$$

Perform the k-means clustering algorithm setting $k = 2$ by using x_1 and x_2 as your initial means.

Solution. If we draw the points in the plane, we can guess that the optimal solution for two clusters will be $\{x_1, x_2, x_3\}$ and $\{x_4, x_5\}$.



Let us now apply our algorithm starting with $m_1^{(1)} = x_1$ and $m_2^{(1)} = x_2$. We have $S_1^{(1)} = \{x_1, x_3, x_4\}$ and $S_2^{(1)} = \{x_2, x_5\}$, which gives $m_1^{(1)} = (-\frac{1}{3}, \frac{1}{3})$ and $m_2^{(2)} = (0, 2)$.



We have $S_1^{(2)} = \{x_1, x_3, x_4\} = S_1^{(1)}$ and $S_2^{(2)} = \{x_2, x_5\} = S_2^{(1)}$, so we stop here. The solution we obtained is not a global optimal, which we could have reached if we had chosen $m_1^{(1)} = x_1$ and $m_2^{(1)} = x_4$ for example. The global optimal solution (for two clusters) can indeed by computed to be $S_1 = \{x_1, x_3, x_4\}$ and $S_2 = \{x_2, x_5\}$ with respective means $m_1 = (-\frac{7}{3}, \frac{4}{3})$ and $m_2 = (3, 0.5)$. ■

Exercise 7.

Let X be the set of points in Question 6. Calculate the relative distance between $\{x_1, x_2, x_3\}$ and $\{x_4, x_5\}$.

Solution. We have $S_1 = \{x_1, x_2, x_3\}$ and $S_2 = \{x_4, x_5\}$. The relative distance between S_1 and S_2 is given by:

$$\begin{aligned}
\mathrm{RD}(S_1 \,||\, S_2) &= \frac{1}{|S_1| \cdot |S_2|} \sum_{x \in S_1, y \in S_2} \mathrm{RD}(x \,||\, y) \\
&= \frac{1}{6}\big(\mathrm{RD}(x_1 \,||\, x_4) + \mathrm{RD}(x_1 \,||\, x_5) + \mathrm{RD}(x_2 \,||\, x_4) + \mathrm{RD}(x_2 \,||\, x_5) \\
&\qquad + \mathrm{RD}(x_3 \,||\, x_4) + \mathrm{RD}(x_3 \,||\, x_5)\big),
\end{aligned}$$

where $\mathrm{RD}(x \,||\, y) = d(x,y) - \frac{1}{n}\sum_{z\in X} d(x,y) - d(x,z)$. Using the table from Question 6 we find:

$$\begin{aligned}
\mathrm{RD}(x_1 \,||\, x_4) &= 2\sqrt{10} - \frac{1}{5}(1+2+2\sqrt{10}+\sqrt{37})\\
\mathrm{RD}(x_1 \,||\, x_5) &= \sqrt{37} - \frac{1}{5}(1+2+2\sqrt{10}+\sqrt{37})\\
\mathrm{RD}(x_2 \,||\, x_4) &= 3\sqrt{5} - \frac{1}{5}(1+\sqrt{5}+3\sqrt{5}+6)\\
\mathrm{RD}(x_2 \,||\, x_5) &= 6 - \frac{1}{5}(1+\sqrt{5}+3\sqrt{5}+6)\\
\mathrm{RD}(x_3 \,||\, x_4) &= 2\sqrt{5} - \frac{1}{5}(2+\sqrt{5}+2\sqrt{5}+\sqrt{17})\\
\mathrm{RD}(x_3 \,||\, x_5) &= \sqrt{17} - \frac{1}{5}(2+\sqrt{5}+2\sqrt{5}+\sqrt{17}).
\end{aligned}$$

Finally,

$$\mathrm{RD}(S_1 \,||\, S_2) = \frac{1}{30}(6+11\sqrt{5}+6\sqrt{10}+3\sqrt{17}+3\sqrt{37}). \qquad \blacksquare$$

Exercise 8.

Let $\mathcal{C}[0,1]$ denote the metric space of continuous real-valued functions on the closed interval $[0,1]$ equipped with the d_{sup} metric. Show that the function $I\colon \mathcal{C}[0,1] \to \mathbf{R}$ defined by $I(f) = \int_0^1 f(x)\,\mathrm{d}x$ is continuous.

Solution. We will prove that using the ε-δ definition of continuity. Suppose that $d_{\text{sup}}(f,g) < \delta$ for some δ. This implies that $|f(x) - g(x)| < \delta$ for all $x \in [0,1]$ and hence that

$$\begin{aligned}
d_{\text{sup}}(I(f), I(g)) &= \sup_{x\in[0,1]} \left| \int_0^1 f(t) - g(t)\,\mathrm{d}t \right|\\
&\leqslant \sup_{x\in[0,1]} \int |f(t) - g(t)|\,\mathrm{d}t\\
&< \sup_{x\in[0,1]} \int \delta\,\mathrm{d}t = \sup_{x\in[0,1]} \delta = \delta
\end{aligned}$$

We just proved that for any $f \in \mathscr{C}[0,1]$, I is continuous at f (by taking $\delta = \varepsilon$). $\blacksquare$

Exercise 9.

Let $g \in \mathcal{C}[0,1]$ and let $F\colon \mathcal{C}[0,1] \to \mathcal{C}[0,1]$ be defined by

$$F(y)(x) = g(x) + \frac{1}{2}\int_0^1 \sin(xt)y(t)\,\mathrm{d}t.$$

Show that F is a contraction mapping and deduce that the equation $y = F(y)$ has a unique solution in $\mathcal{C}[0,1]$.

Solution. First of all, the continuity of y and g on $[0,1]$ (and of sin) implies that $F(y)$ is also continuous. So F is a well-defined self-map. Now, let y_1 and y_2 be two functions in $\mathscr{C}[0,1]$. We have

$$d_{sup}(F(y_1), F(y_2)) = \sup\left\{ \frac{1}{2}\int_0^1 \sin(xt)|y_1(t) - y_2(t)|\,\mathrm{d}t \;\middle|\; x \in [0,1] \right\},$$

where we used the fact that $\sin(xt) \geqslant 0$ for all x and t in $[0,1]$. Since we also have $\sin(xt) \leqslant 1$, we conclude that

$$\begin{aligned} d_{sup}(F(y_1), F(y_2)) &\leqslant \frac{1}{2} d_{\sup}(y_1, y_2) \cdot \sup\left\{ \int_0^1 \sin(xt)\,\mathrm{d}t \;\middle|\; x \in [0,1] \right\} \\ &\leqslant \frac{1}{2} d_{\sup}(y_1, y_2), \end{aligned}$$

which shows that F is a contraction (with Lipschitz constant $\frac{1}{2}$).

Since $\mathscr{C}[0,1]$ is complete, we can use the Banach's Contraction Mapping Theorem to conclude that F has a unique fixed point. In other words, the equation $y(x) = g(x) + \frac{1}{2}\int_0^1 \sin(xt)y(t)\,\mathrm{d}t$ has a unique solution in $\mathscr{C}[0,1]$. ■